

Participation, rearrangement, and fragmentation

36

Connections

➡ Building on

- Nucleophilic substitution at saturated carbon [ch15](#)
- Conformational analysis [ch16](#)
- Elimination reactions [ch17](#)
- Electrophilic aromatic substitution [ch21](#)
- Controlling stereochemistry [ch14](#), [ch32](#), & [ch33](#)
- Main group chemistry [ch27](#)
- Stereoelectronics [ch31](#)
- Sigmatropic rearrangements [ch35](#)

Arriving at

- Participation: nucleophiles that are already part of the molecule
- Participation may mean rearrangement
- Participating groups can have lone pairs or π electrons
- Carbocations often rearrange by alkyl migration
- How to work out the mechanism of a rearrangement
- Ring expansion by rearrangement
- Using rearrangements in synthesis
- Insertion of O, N, or C next to a ketone
- How fragmentation splits molecules into three pieces by C—C bond cleavage
- Controlling rearrangements and fragmentations
- Control of fragmentations by stereochemistry

➡ Looking forward to

- Carbene chemistry [ch38](#)
- Determination of mechanism [ch39](#)
- The chemistry of life [ch42](#)

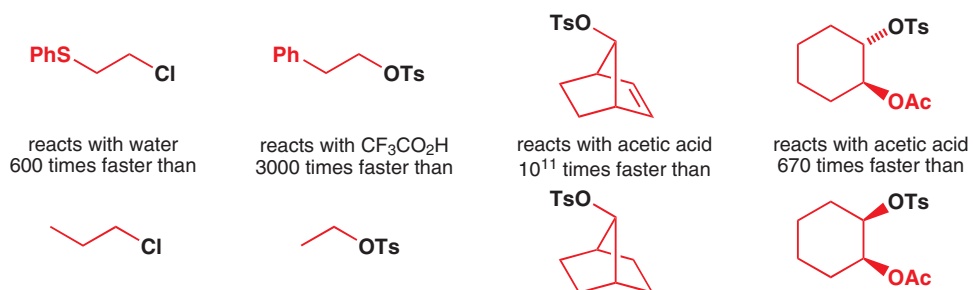
The last two chapters introduced **pericyclic reactions**, and the next one will cover reactions of **radicals**. Together with the **ionic reactions** which have been the subject of most of this book, these three classes cover all organic mechanisms. But before we move on to consider radicals, we need to fill a gap in our coverage of ionic reactions. You have met the most important types of ionic reactions—additions, substitutions, and eliminations. But two remain and they are closely related. In **rearrangements** the molecule changes its carbon skeleton and in **fragmentations** the carbon skeleton splits into pieces. We lead up to these types of reaction by looking at a phenomenon known as **participation**.

Neighbouring groups can accelerate substitution reactions

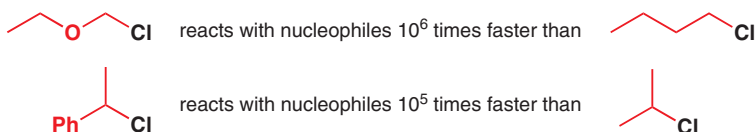
Compare the rates of the following substitution reactions. Each is a substitution of the leaving group (OTs or Cl) by solvent, known as a solvolysis.

■ A solvolysis was defined in Chapter 15 as 'a reaction in which the solvent is also the nucleophile'.

Online support. The icon  in the margin indicates that accompanying interactive resources are provided online to help your understanding: just type www.chemtube3d.com/clayden/123 into your browser, replacing **123** with the number of the page where you see the icon. For pages linking to more than one resource, type **123-1**, **123-2** etc. (replacing **123** with the page number) for access to successive links.

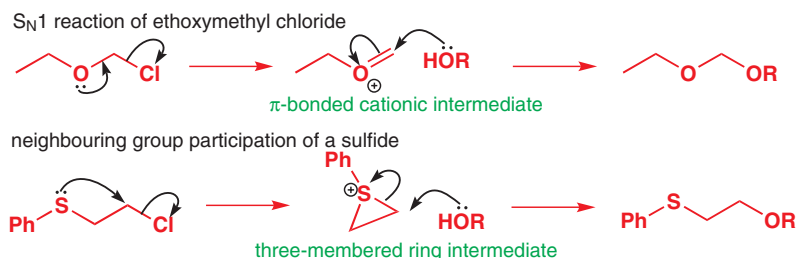


Nearby groups can evidently increase the rate of substitution reactions significantly. Now, you may be thinking back to Chapter 15 and saying ‘yes, yes, we know that’—when we were discussing the mechanisms of substitution reactions we pointed out that a cation-stabilizing group at the reaction centre makes S_N1 reactions very fast, for example:



In the four examples above, though, it is not at the reaction centre itself that the functional groups change but at the carbon *next* to the reaction centre, and we call these groups *neighbouring groups*. The mechanism by which they speed up the reactions is known as *neighbouring group participation*. Compare the reaction of this ether and this sulfide with an alcohol.

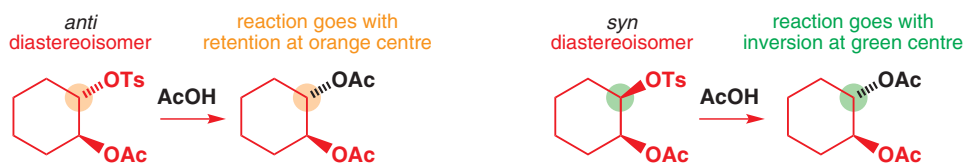
■ Neighbouring group participation is occasionally called **anchimeric assistance** (Greek *anchi* = neighbouring; *mer* = part).



In both cases, ionization of the starting material is assisted by the lone pair of an electron-rich functional group. The ether in the first example assists by forming a π bond, the sulfide assists by forming a σ bond in a three-membered ring, and a common feature of all mechanisms involving neighbouring group participation is the formation of a cyclic intermediate.

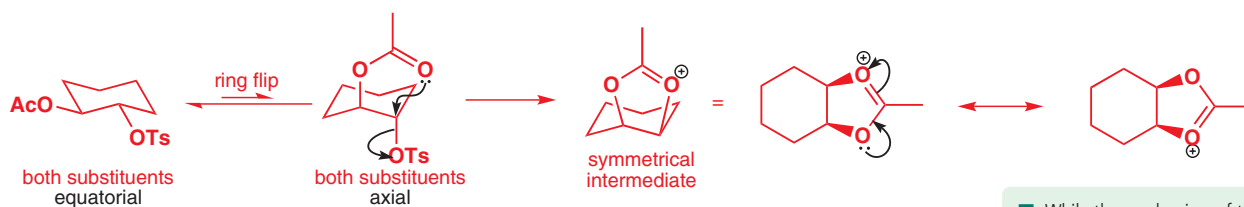
Stereochemistry can indicate neighbouring group participation

How do we know that neighbouring group participation is taking place? Well, the first bit of evidence is the *increase in rate*. The neighbouring groups become involved only if they can increase the rate of the substitution reaction—otherwise the mechanism will just follow the ordinary S_N2 pathway. But more important information comes from reactions where stereochemistry is involved, and one of these is the last of the four examples at the start of the chapter. Here it is again in more detail. Not only does the first of these reactions go faster than the second—its stereochemical course is different too.



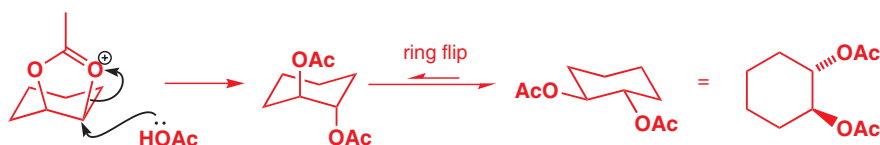
Although one starting material has *syn* and the other *anti* stereochemistry, the products have the same (*anti*) stereochemistry one substitution goes with retention and one goes with

inversion. Again, neighbouring group participation is the reason. To explain this, we should first draw the six-membered rings in their real conformation. For the *anti* compound, both substituents can be equatorial. However, not much can happen in this conformation—but, if we allow the ring to flip, you can see immediately that the acetate substituent is ideally placed to participate in the departure of the tosylate group.



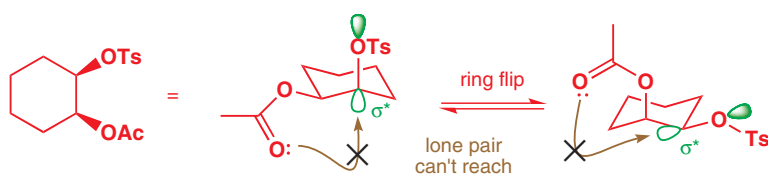
What results is an entirely symmetrical intermediate—the positive charge on one of the oxygens is, of course, delocalized over both of them. The intramolecular S_N2 reaction takes place with inversion, as required by the orbitals, so now the junction of the two rings is *cis*.

The next step is attack of acetic acid on the intermediate. This is another S_N2 reaction, which also proceeds with inversion and gives back a *trans* product.

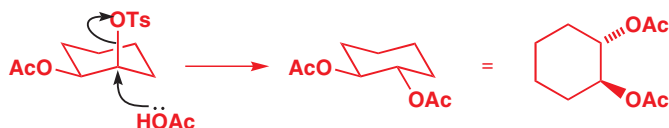


Overall, we have *retention* of stereochemistry. As you know, S_N2 reactions go with inversion and S_N1 reactions with loss of stereochemical information, so this result is possible only if we have two sequential S_N2 reactions taking place—in other words neighbouring group participation.

Why, then, does the other diastereoisomer react with inversion of stereochemistry? Well, try drawing the mechanism for intramolecular displacement of the tosyl group. Whether you put the tosylate or the acetate group equatorial doesn't matter; there is no way in which the acetate oxygen's lone pair can reach the σ^* orbital of the tosylate C—O bond.



Neighbouring group participation is impossible, and substitution goes simply by intermolecular displacement of OTs by AcOH. Just one S_N2 step means overall inversion of configuration, and no participation means a slower reaction.



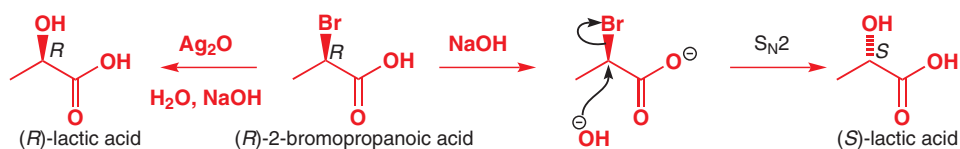
Retention of configuration is an indication of neighbouring group participation

Enantiomerically pure (*R*)-2-bromopropanoic acid reacts with concentrated sodium hydroxide to give (*S*)-lactic acid. The reaction goes with inversion and is a typical S_N2 reaction—and a good one too, since the reaction centre is adjacent to a carbonyl group (see Chapter 15). If, on the other hand, the reaction is run using Ag_2O and a low concentration of sodium hydroxide, (*R*)-lactic acid is obtained—there is overall *retention* of stereochemistry.

➡ If you are unsure what we are talking about, go back and read Chapter 16 now!

■ While the mechanism of this first step of the substitution reaction is S_N2 in appearance—a nucleophile (the acetate group) arrives just as a leaving group (the tosylate group) departs—it is also, of course, only unimolecular.

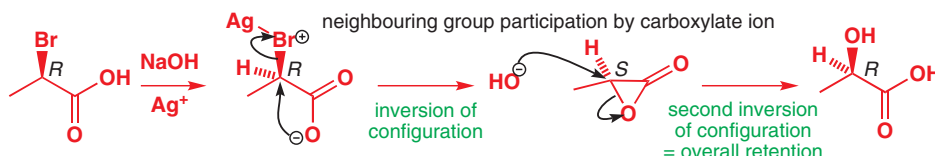
🔗 Interactive mechanism for unexpected retention of stereochemistry



■ Lactones (that is, cyclic esters) don't usually react with hydroxide by this mechanism, and you might expect this intermediate (which is a cyclic ester) to hydrolyse by attack of hydroxide at the C=O group. You might like to think about why this doesn't happen in this case.

Interactive mechanism for α -lactone formation

Nucleophilic substitution reactions that go with retention of stereochemistry are rather rare and mostly go through two successive inversions with neighbouring group participation, like the example you saw in the last section. This time the neighbouring group is carboxylate: the silver oxide is important because it encourages the ionization of the starting material by acting as a halogen-selective Lewis acid. A three-membered ring intermediate forms, which then gets opened by hydroxide in a second $\text{S}_{\text{N}}2$ step.



● Retention suggests participation

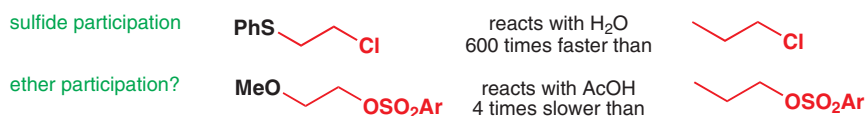
If you see a substitution reaction at a stereogenic saturated carbon atom that goes with retention of stereochemistry, look for neighbouring group participation!

Why does the carboxylate group participate only at low HO^- concentration and in the presence of Ag^+ ? You can think of the situation in these two reactions in terms of the factors that favour $\text{S}_{\text{N}}1$ and $\text{S}_{\text{N}}2$ reactions. In the first, we have conditions suited to an $\text{S}_{\text{N}}2$ reaction: a very good nucleophile (HO^-) and a good leaving group (Br^-). Improve the leaving group by adding Ag^+ (Ag^+ assists Br^- 's departure much as H^+ assists the departure of OH^- by allowing it to leave as H_2O) and worsen the nucleophile (H_2O instead of HO^- , of which there is now only a low concentration), and we have the sorts of conditions that *would* favour an $\text{S}_{\text{N}}1$ reaction. The trouble is, without neighbouring group participation, the cation here would be rather unstable—right next to a carbonyl group. The carboxylate saves the day by participating in the departure of the Br^- and forming the lactone. The key thing to remember is that a reaction always goes by the mechanism with the fastest rate.

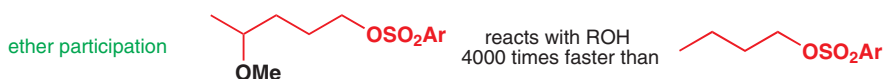
● Neighbouring groups participate only if they speed up the reaction.

What sorts of groups can participate?

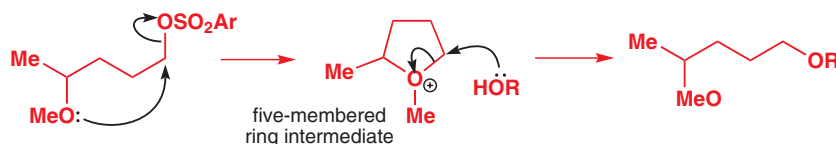
You've already met the most important ones—sulfides, esters, carboxylates. Ethers and amines (you will see some of these shortly) can also assist substitution reactions through neighbouring group participation. The important thing that they have in common is an electron-rich heteroatom with a lone pair that can be used to form the cyclic intermediate. Sulfides are rather better than ethers—this sulfide reacts with water much faster than $n\text{-PrCl}$ but the ether reacts with acetic acid four times more *slowly* than $n\text{-PrOSO}_2\text{Ar}$.



The OMe group slows the reaction down just because it is electronegative more than it accelerates it by participation. A more distant OMe group can participate: this 4-MeO alkyl sulfonate reacts with alcohols 4000 times faster than the $n\text{-Bu}$ sulfonate.



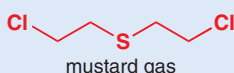
Again neighbouring group participation is involved, but this time through a five- rather than a three-membered ring. Participation is most commonly through three- and five-membered rings, less often six-membered ones, and very rarely four- or more than seven-membered ones.



■ Why these ring sizes? Well, the underlying reasons are the same as those we discussed in Chapter 31 when we talked about the kinetics (rates) of formation and thermodynamics (stability) of different ring sizes: three- and five-membered rings form particularly rapidly in any reaction.

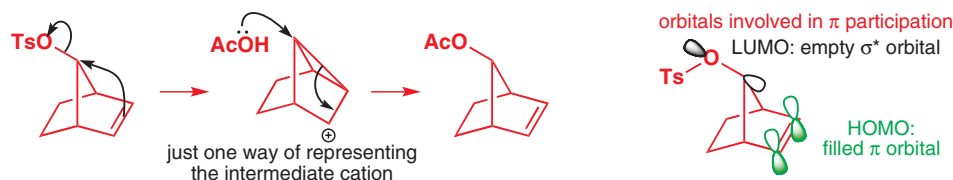
Mustard gas

Participation of sulfides through three-membered rings was used to gruesome effect in the development of mustard gas during the Second World War. Mustard gas itself owes its toxicity to the neighbouring group participation of sulfur, which accelerates its alkylation reactions.



Not all participating groups have lone pairs

Another of the four examples we started with shows that even the π electrons of a $C=C$ double bond can participate. Retention of stereochemistry in the product (the starting tosylate and product acetate are both *anti* to the double bond) and the extremely fast reaction (10^{11} times that of the saturated analogue) are tell-tale signs of neighbouring group participation.

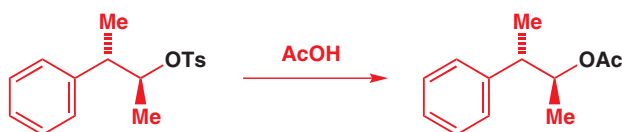


What is the structure of the intermediate?

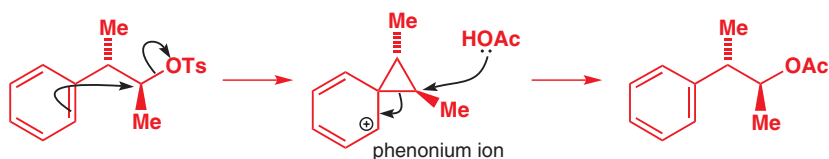
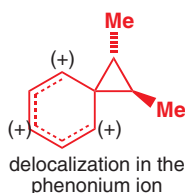
During the 1950s and 1960s, this sort of question provoked a prolonged and acrimonious debate, which we have no intention of stirring up, and all we will do is point out that the intermediate in this reaction is not fully represented by the structure we have here: it is symmetrical and could be represented by two structures with three-membered rings or by a delocalized structure in which two electrons are shared between three atoms. The difference need not concern us.



Finally, an example with a neighbouring phenyl group. Participation is suggested by the retention of relative stereochemistry.

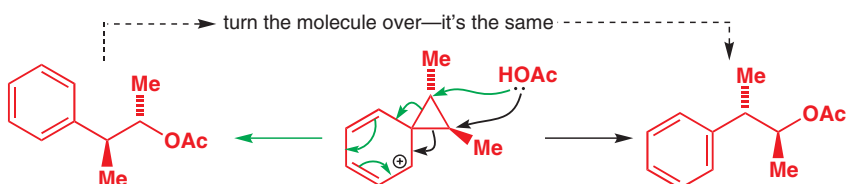


Again, π electrons are involved, but the reaction is now electrophilic aromatic substitution (Chapter 21) rather like an intramolecular Friedel–Crafts alkylation with a delocalized intermediate often termed a phenonium ion.



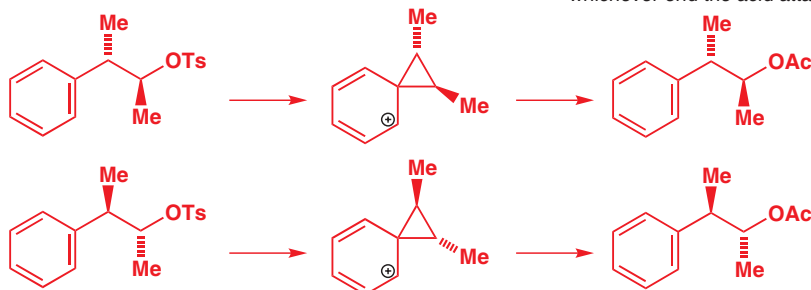
More stereochemical consequences of neighbouring group participation

The phenonium ion is symmetrical. The acetic acid can attack either atom in the three-membered ring to give the same product.



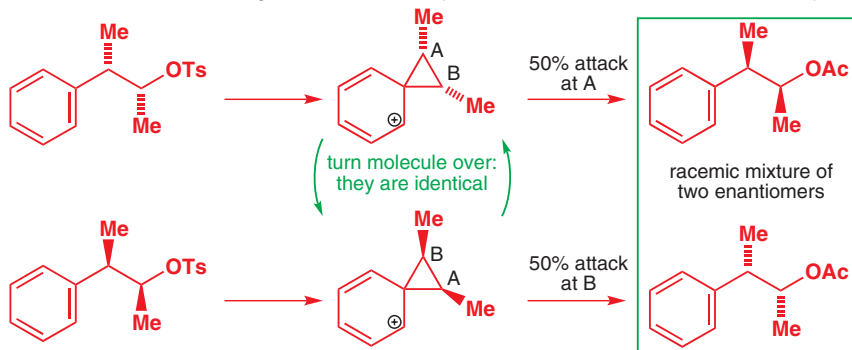
The phenonium ion is nonetheless still chiral, since it has an axis (and not a plane or centre) of symmetry, so if we use an enantiomerically pure starting material we get an enantiomerically pure product.

from this enantiomer of tosylate . . . we get this phenonium ion . . . and this enantiomer of product whichever end the acid attacks



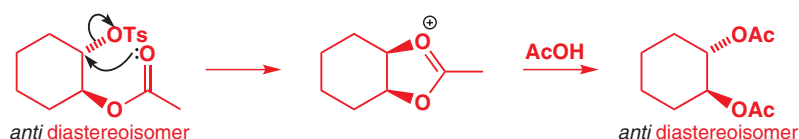
Not so with the other diastereoisomer of this compound! Now, the phenonium ion is symmetrical with a plane of symmetry—it is therefore achiral, and the same whichever enantiomer we start from. Attack on each end of the phenonium ion gives a different enantiomer, so whichever enantiomer of starting material we use we get the same racemic mixture of products. You can compare this reaction with the loss of stereochemical information that occurs during an S_N1 reaction of enantiomerically pure compounds. Both reactions pass through an achiral intermediate.

from either enantiomer . . . we get the same achiral phenonium ion . . . and therefore racemic product

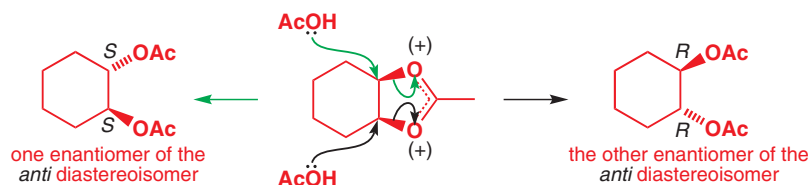


There is a subtlety here that you should not overlook and that makes this study, which was carried out by Cram in 1949, exceedingly elegant. Reactions of both of these diastereoisomers are stereospecific: the *relative* stereochemistry of the products depends on the *relative* stereochemistry of the starting materials. Yet, while the absolute stereochemistry of the starting materials is retained in one case (we get a single enantiomer of a single diastereoisomer), it is lost in the other (we get a racemic mixture of both enantiomers of a single diastereoisomer). These are important distinctions, and if you are in any doubt about these terms, re-read Chapters 14 and 33. Donald Cram (1919–2001) of UCLA was awarded the Nobel prize in 1987 jointly with Jean-Marie Lehn (1939–) of Strasbourg and Paris, and Charles Pedersen (a Norwegian born in Korea in 1904) of DuPont for 'their development and use of molecules with structure-specific interactions of high selectivity'.

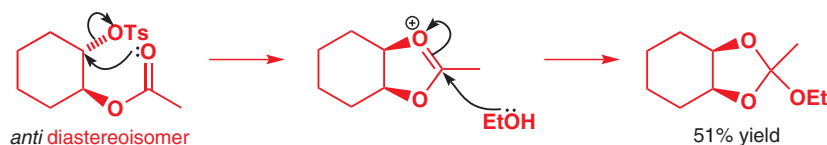
The same loss of absolute stereochemical information (but retention of relative stereochemistry) occurs in another reaction that you met at the start of this chapter. We then emphasized two features: the acceleration in rate and the retention of stereochemistry.



The intermediate oxonium ion is delocalized and achiral. If a single enantiomer of the starting material is used, racemic product is formed through this achiral intermediate. Attack at one carbon atom gives one enantiomer; attack at the other gives the mirror image.

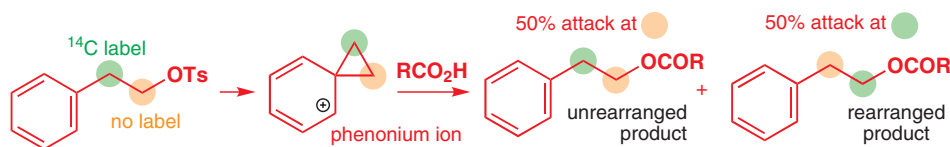


In this case the neighbouring group can be caught in the act—when the rearrangement is carried out in ethanol, the intermediate is trapped by attack at the central carbon atom. It is as though someone switched the light on while the acetate's fingers were in the biscuit tin. The product is an orthoester and is achiral too. This chemistry should remind you of the formation of acetals, as described in Chapter 11.



Rearrangements occur when a participating group ends up bonded to a different atom

Because the intermediates in these examples are symmetrical, 50% of the time one substituent ends up moving from one carbon atom to another during the reaction. This is clearer in the following example: the starting material is prepared such that the carbon atom carrying the phenyl group is an unusual isotope—carbon-14. This doesn't affect the chemistry, but means that the two carbon atoms are easily distinguishable. Reacting the compound with trifluoroacetic acid scrambles the label between the two positions: the intermediate is symmetrical and, in the 50% of reactions with the nucleophile that take place at the labelled carbon atom, the phenyl ends up migrating to the unlabelled carbon atom in a rearrangement reaction.

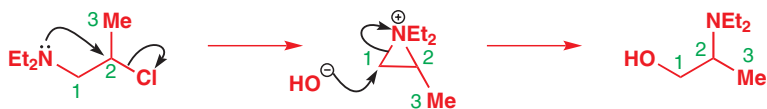


Now, consider this substitution reaction, in which OH replaces Cl but with a change in the molecular structure. The substitution goes with complete rearrangement—the amine ends up attached to a different carbon atom. We can easily see why if we look at the mechanism. The reaction starts off looking like a neighbouring group participation of the sort you are now familiar with (the carbon atoms are numbered for identification).

■ Labelling an atom with an unusual isotope is a standard way to probe the details of a reaction. Radioactive ^3H (tritium) or ^{14}C used to be used but, with the advent of high-field NMR, non-radioactive ^2H (deuterium) and ^{13}C are more versatile and less hazardous. These methods are examined more thoroughly in Chapter 39.

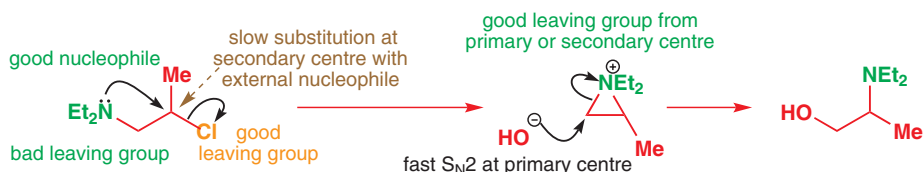


The intermediate is an aziridinium ion (aziridines are three-membered rings containing nitrogen—the nitrogen analogues of epoxides). The hydroxide ion chooses to attack only the less hindered terminal carbon 1, and a rearrangement results—the amine has migrated from carbon 1 to carbon 2.



Interactive mechanism for migration of participating group

We should just pause here for a moment to consider why this rearrangement works. We start with a secondary alkyl chloride that contains a very bad leaving group (Et_2N) and a good one (Cl^-)—but the good one is hard for HO^- to displace because it is at a secondary centre (remember—secondary alkyl halides are slow to react by $\text{S}_{\text{N}}1$ or $\text{S}_{\text{N}}2$). But the NEt_2 can participate to make an aziridinium intermediate—now there is a good leaving group (RNEt_2 without the negative charge) at the primary as well as the secondary carbon, so HO^- does a fast $\text{S}_{\text{N}}2$ reaction at the primary carbon.

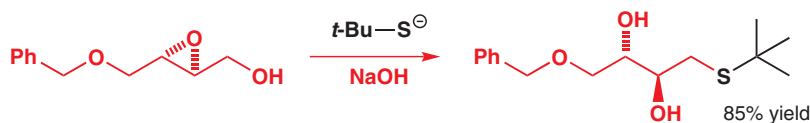


Another way to look at this reaction is to see that the good internal nucleophile Et_2N will compete successfully for the electrophile with the external nucleophile HO^- . Intramolecular reactions are usually faster than bimolecular reactions.

- Intramolecular reactions (including participation of a neighbouring group) that give three-, five-, or six-membered rings are usually faster than intermolecular reactions.

The Payne rearrangement

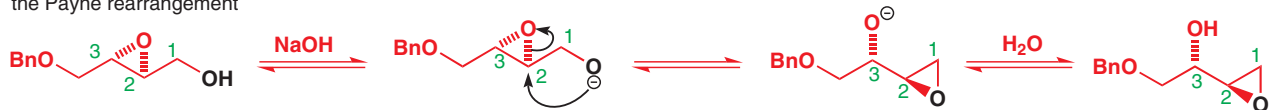
The reaction of an epoxy alcohol in base does not always give the expected product.



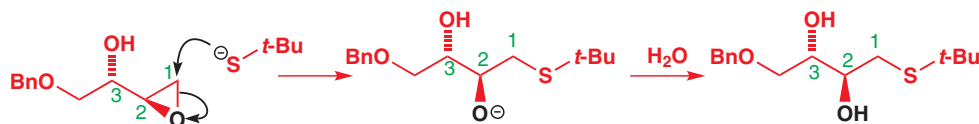
The thiolate nucleophile has not opened the epoxide directly, but instead *appears* to have displaced HO^- —a very bad leaving group. Almost no nucleophile will displace OH^- , so we need an alternative explanation. This comes in the form of another rearrangement, this time involving oxygen, but otherwise rather similar to the ones you have just met. Again, our epoxide, although reactive as an electrophile, suffers from being secondary at both electrophilic centres. $t\text{-BuS}^-$ is a bulky nucleophile, so direct attack on the epoxide is slow. Instead, under the basic conditions of the reaction, the neighbouring alkoxide group attacks intramolecularly to make a new, rearranged epoxy alcohol. This is called the Payne rearrangement.

Interactive mechanism for Payne rearrangement

the Payne rearrangement



Now we do have a reactive, primary electrophilic site, which undergoes an S_N2 reaction with the $t\text{-BuS}^-$ under the conditions of the rearrangement. Notice how the black OH, which started on the carbon labelled 1, has ended up on carbon 2.



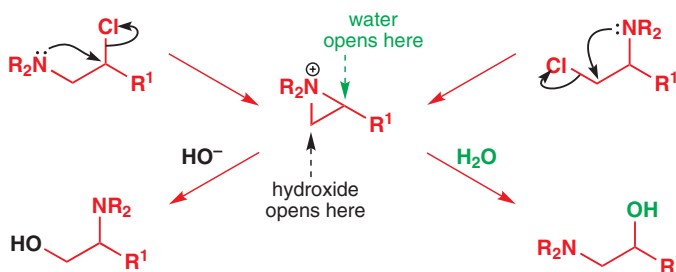
The direction of rearrangement can depend on the nucleophile

Compare these reactions: you saw the first on p. 938 but the second is new.

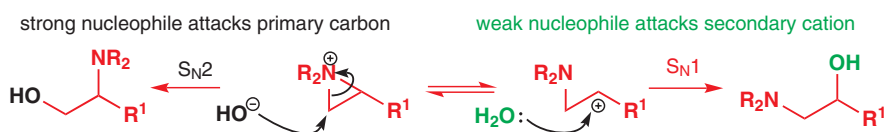


In the first reaction, the amine migrates from the primary to the secondary position; in the other from secondary to primary. Both go through very similar aziridinium intermediates, so the difference must be due to the regioselectivity with which this aziridinium ion opens in each case.

The only important difference is the nucleophile used in the reaction. Hydroxide opens the aziridinium at the less hindered end; water opens the aziridinium ion at the more hindered (more substituted) end. Why?



We can think of the aziridinium ion as a compound containing two alternative leaving groups—one from a primary centre and one from a secondary one. Primary centres can take part in fast S_N2 reactions, but cannot undergo S_N1 . Secondary centres can undergo either S_N1 or S_N2 reactions, but, in general, do neither very well. Now, the rate of an S_N2 reaction depends on the nucleophile, so a good nucleophile (like HO^-) can do fast S_N2 reactions, while a bad one (like H_2O) cannot. The fastest reaction HO^- can do then is S_N2 at the primary centre (remember: you see only the reaction that goes by the fastest mechanism). Water, on the other hand, takes part only reluctantly in substitution reactions—but this does not matter if they are S_N1 reactions because their rates are independent of nucleophile. H_2O waits until the leaving group has left of its own accord to give a cation, which rapidly grabs *any* nucleophile—water will do just as well as HO^- . This can happen *only* at the secondary centre because the primary cation is too unstable to form.



All the rearrangements you have met so far occurred during substitution reactions. All happened because reaction *with* rearrangement is faster than reaction *without* rearrangement—in other words, rearrangement occurs because of a kinetic preference for the rearrangement pathway. You could see these reactions as ‘special case’ examples of neighbouring group participation—in both participation and rearrangement the neighbouring group speeds up the

■ When a group migrates from a primary to a secondary carbon, we say the rearrangement has a primary **migration origin** and a secondary **migration terminus**. The migrating group moves from the migration origin to the migration terminus.

Interactive mechanism showing the effect of different nucleophiles

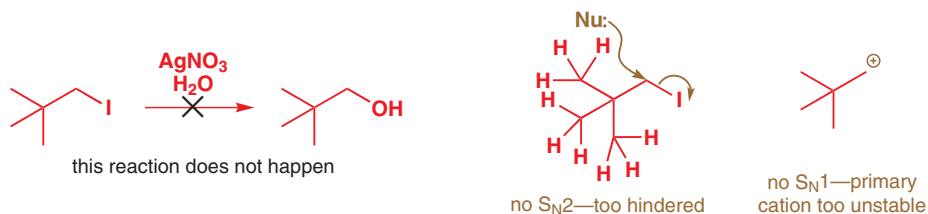
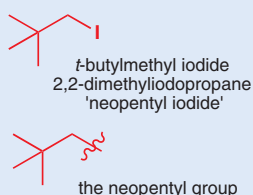
reaction, but in rearrangement reactions the neighbouring group gets rather more than it bargained for, and ends up elsewhere in the molecule. Both proceed through a cyclic transition state or intermediate, and it is simply the way in which that transition state or intermediate collapses that determines whether rearrangement occurs.

Rearrangement can involve migration of alkyl groups

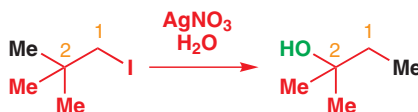
This example is a nucleophilic substitution under conditions (Ag^+ , H_2O) designed to encourage $\text{S}_{\text{N}}1$ reactions (excellent leaving group, poor nucleophile). First of all, this is what does not happen (and indeed without Ag^+ *nothing happens at all*).

Neopentyl

The *t*-butylmethyl group is also called 'neopentyl'.

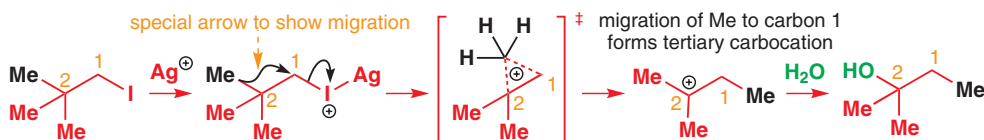


Compounds like this, with a *t*-butyl group next to the electrophilic centre, are notoriously slow to undergo substitution reactions. They can't do $\text{S}_{\text{N}}2$, they are too hindered; they can't do $\text{S}_{\text{N}}1$, the cation you would get is primary. In fact, a rearrangement occurs. One of the methyl groups moves ('migrates') from carbon 2 to carbon 1, the new OH group taking its place at carbon 2.



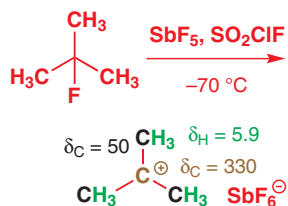
How has this happened? Well, firstly, our principle (p. 934) tells us that it has happened because $\text{S}_{\text{N}}1$ and $\text{S}_{\text{N}}2$ are both so slow that this new rearrangement mechanism is faster than either. Adding Ag^+ makes I^- desperate to leave, but unassisted this would mean the formation of a primary carbocation. The molecule does the only thing it can to stop this happening and uses the electrons in an adjacent C—C bond to assist the departure of I^- . Having participated, the methyl group continues to migrate to carbon 1 because by doing so it allows the formation of a stable tertiary carbocation, which then captures water in a step reminiscent of the second half of an $\text{S}_{\text{N}}1$ reaction. Note the cyclic transition state where the migrating group is partially bonded to two carbon atoms.

■ Some of the cyclic species you have seen so far (aziridinium ions, epoxides) are intermediates; the intermediate cyclic cation here is probably only a transition state.



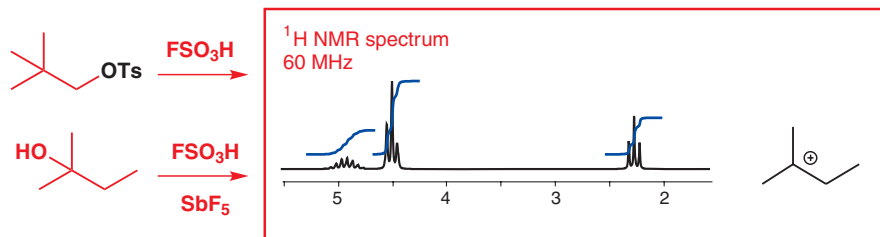
In the migration step we used a slightly unusually S-shaped curly arrow to represent the movement of a group (Me) along a bond taking its bonding electrons with it. We shall use this type of arrow when a group migrates from one atom to another during a rearrangement.

Carbocations readily rearrange



In Chapter 15 we showed you that it is possible to run the NMR spectra of carbocations by using a polar but non-nucleophilic solvent such as liquid SO_2 or SOClF . Treating an alkyl halide RX with the powerful Lewis acid SbF_5 under these conditions gives a solution of carbocation: the carbocation reacts neither with solvent nor the SbF_5X^- counterion because neither is nucleophilic. We know, for example, that the chemical shifts in both the ^{13}C and ^1H NMR spectra of the *t*-butyl cation are very large, particularly the ^{13}C shift at the positively charged centre.

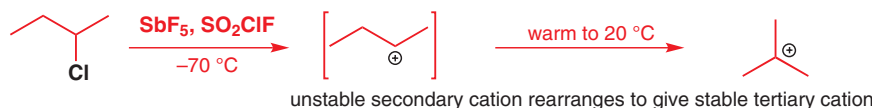
NMR can be used to follow the course of rearrangement reactions involving carbocations too. We can illustrate this with an experiment that tries to make the neopentyl cation by the substitution reaction you have just seen. This time the starting material and solvent are slightly different, but the outcome is nonetheless most revealing. Dissolving neopentyl tosylate in fluorosulfonic acid (a strong, non-nucleophilic acid) at -77°C gives a 77% yield of a cation whose spectrum is shown below. Assigning the peaks is not hard once you know that the same spectrum is obtained when 2,2-dimethyl-2-butanol is dissolved in fluorosulfonic acid with SbF_5 added.



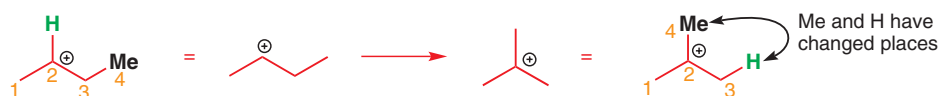
■ Notice how the methyl groups appear as triplets due to coupling to CH_2 through the empty p orbital.

Clearly, the spectrum is the tertiary 2-methylbutyl cation and the neopentyl cation never saw the light of day. The reaction is the same rearrangement that you saw in the substitution reaction of neopentyl iodide, but here the rate of rearrangement can be measured and it is extremely fast. Neopentyl tosylate reacts to form a cation under these conditions about 10^4 times as fast as ethyl tosylate, even though both tosylates are primary. This massive rate difference shows that if migration of an alkyl group can allow rearrangement to a more stable carbocation, it will happen, and happen rapidly.

Primary cations can never be observed by NMR—they are too unstable. But secondary cations can, provided the temperature is kept low enough. *sec*-Butyl chloride in SO_2ClF at -78°C gives a stable, observable cation. But, as the cation is warmed up, it rearranges to the *t*-butyl cation. Now this rearrangement truly is a carbocation rearrangement: the starting material is an observable carbocation and so is the product, and we should just look at the mechanism in a little more detail.



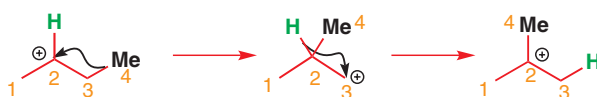
With rearrangements like this it is best to number the C atoms so you can see clearly what moves where. If we do this, we see that the methyl group we have labelled 4 and the H on C3 have changed places. (Note that C3 starts off as a CH_2 group and ends up as CH_3 .)



● Top tip for rearrangements

Number the carbon atoms in the starting material and product before you try to work out the mechanism.

Using the sort of arrows we introduced on p. 940, we can draw a mechanism for this in which first the Me migrates, and then the hydride. We say hydride migration rather than *hydrogen* (or *proton*) because the H atom migrates *with* its pair of electrons.



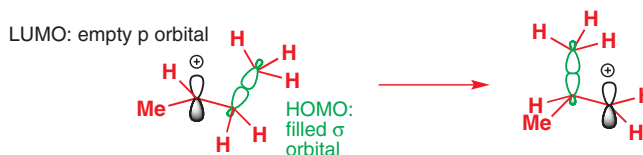
As these rearrangements are a new type of reaction, we should just spend a moment looking at the molecular orbitals that are involved. For the first step, migration of the methyl group,

■ In fact, all seven possible isomers of pentyl alcohol ($\text{C}_5\text{H}_{11}\text{OH}$) give this same spectrum under these conditions at temperatures greater than -30°C .

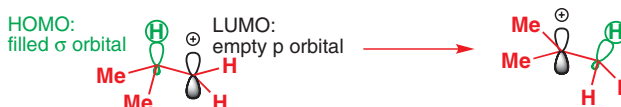
■ The reason we say 'truly is a carbocation rearrangement' here is quite subtle and need not detain us long. We know that a secondary cation is formed in this case because we can see it by NMR; it subsequently rearranges to a tertiary cation. As we can never see primary cations, we don't know that they are ever formed, and the most reasonable explanation for rearrangements of the type you saw on p. 937 is that migration of the alkyl group begins *before* the leaving group is fully gone. This has been proved in a few cases, but we will from now on not distinguish between the two alternatives.

■ You will see why Me has to migrate first if you try drawing the mechanism out with H migrating first instead.

the LUMO must clearly be the empty p orbital of the cation, and the HOMO is the C–C σ bond, which is about to break.

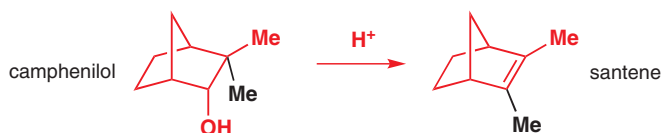


The methyl group can slide smoothly from one orbital to another—there are bonding interactions all the way. The next step, migration of H, is just the same—except that the HOMO is now a C–H σ bond. The methyl migration is thermodynamically unfavourable as it transforms a secondary cation into an unstable primary cation but the hydride migration puts that right as it gives a stable tertiary cation. The whole reaction is under thermodynamic control.

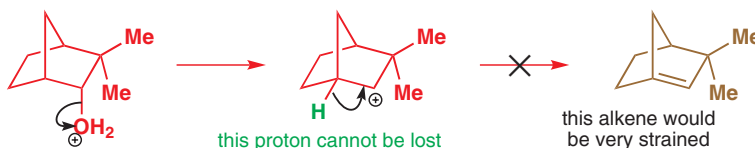


Wagner–Meerwein rearrangements

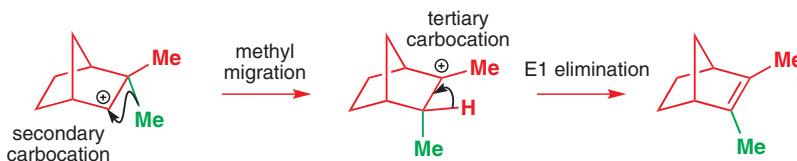
Carbocation rearrangements involving migration of H or alkyl groups don't just happen in NMR machines. They happen during normal reactions too. For example, acid-catalysed dehydration of the natural product camphenilol gives the alkene santene (a key component of the fragrance of sandalwood oil) in a reaction involving migration of a methyl group.



The mechanism shows why the rearrangement happens: the first-formed cation cannot eliminate H^+ in an E1 reaction because loss of the only available proton would give a very strained bridgehead alkene (make a model and see!).



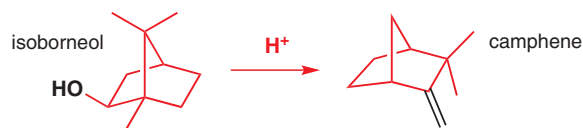
However, migration of a methyl group both stabilizes the cation—it becomes tertiary instead of secondary—and allows E1 elimination of H^+ to take place to give a stable alkene.



➡ The impossibility of bridgehead alkenes (Bredt's rule) was discussed in Chapter 17, p. 389.

🖱 Interactive mechanism for Wagner–Meerwein rearrangements

The migration of an alkyl group to a cationic centre is known as a Wagner–Meerwein rearrangement or Wagner–Meerwein shift, and this migration is, of course, a synthetic manifestation of the rearrangement we have just been looking at in NMR spectra. Wagner–Meerwein shifts have been studied extensively in the class of natural products to which both of these natural products belong—terpenes. For the moment, though, we will just illustrate this type of reaction with one more example—another acid-catalysed dehydration, of isborneol to give camphene.

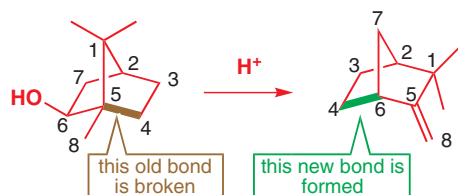


This one *seems* much more complicated—but, in fact, only one alkyl migration is involved. To see what has happened, remember the ‘top tip’—number the carbons. You can number the starting material any way you choose—we’ve started with the gem-dimethyl group because it will be easy to spot in the product. The numbers just follow round the ring, with C8 being the methyl group attached to C5.

Now for the hard bit—we need to work out which carbon in the starting material becomes which carbon in the product. The best thing is just to have a go—mistakes will soon become obvious and you can always try again.

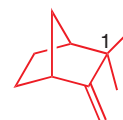
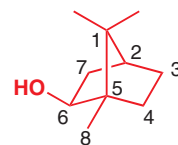
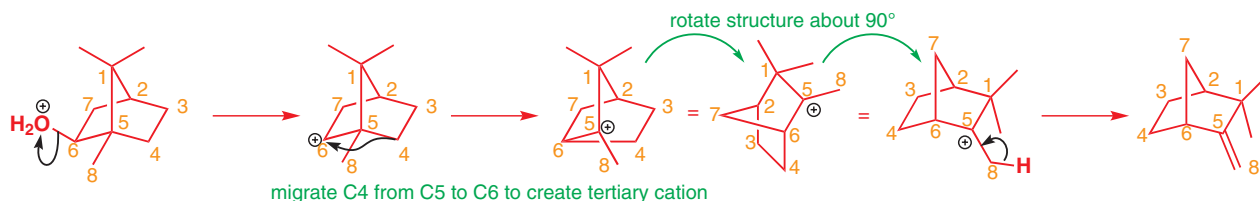
- Use the substituents to help you—some will have changed, but most will be the same or similar, for example C1 is still easy to spot as the carbon carrying the dimethyl group.
- Use connectivity to help you—again, a C–C bond or two may have broken or formed, but most of the C–C bonds in the starting material will be there in the product. C1 and C2 will probably still be next door to one another—C2 was a bridgehead carbon in the starting material, and there is a bridgehead C attached to C1 in the product; assume that’s C2.
- C3 and C4 were unsubstituted carbons in the starting material, and are identifiable in the product too. The other easily spotted atom is C7—an unsubstituted C attached to C2.
- C5, C6, and C8 are harder. We can assume that C8 is the $=CH_2$ carbon—it was a methyl group but perhaps has become involved in an elimination. C5 was attached to C1, C4, C6, and C8: one of the remaining carbons is attached to C1 and C8, so that seems more likely to be C5, which leaves C6 as the bridgehead, attached as before to C7 and C5.

Now we have the whole picture and we can assess what has happened in the reaction—which old bonds have broken and which new bonds have formed.



Numbering the atoms this way identifies the likely point of rearrangement—the only bond broken is between C4 and C5. Instead we have a new one between C4 and C6: C4 appears to have migrated from C5 to C6.

Now for the mechanism. The first step will, of course, be loss of water to generate a secondary cation at C6. The cation is next to a quaternary centre, and migration of any of three bonds could generate a more stable tertiary carbocation. But we know that the new bond in the product is between C4 and C6, so let’s migrate carbon 4. Manipulating the diagrams a bit turns up a structure remarkably similar to our product, and all we need to do is lose a proton from C8.



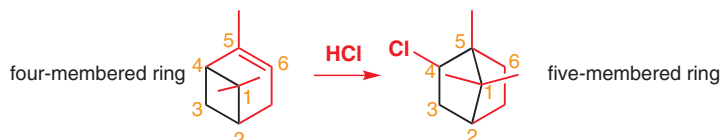
■ If you are observant, you may ask why the alkyl group migrated in this example and not the methyl group, or the other alkyl group—all three possibilities give similar tertiary carbocations. The reason involves the *alignment* of the orbitals involved, which we will discuss at the end of the chapter.

Although migration of an alkyl group that forms part of a ring leads to much more significant changes in structure than simple migration of a methyl group, the reason why it happens is still just the same.

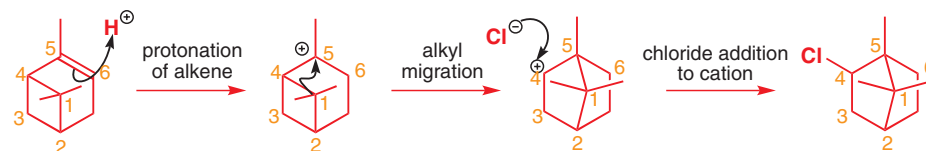
● Alkyl migrations occur in order to make a carbocation more stable.

Ring expansion means rearrangement

‘More stable’ usually means ‘more substituted’, but cations can also be made *more stable* if they become *less strained*. So, for example, four-membered rings adjacent to cations readily rearrange to five-membered rings in order to relieve ring strain.

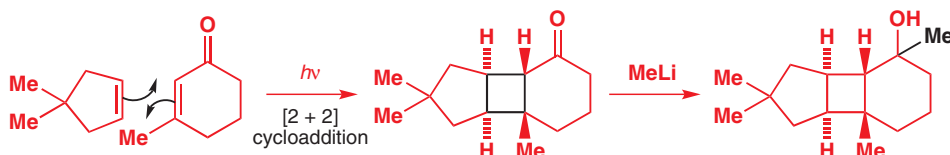


This time the cation is formed by protonation of an alkene, not departure of a leaving group, but writing a mechanism should now be a straightforward matter to you.

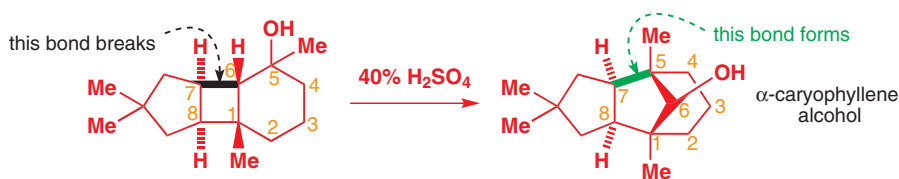


Interactive mechanism for cation-mediated ring expansion

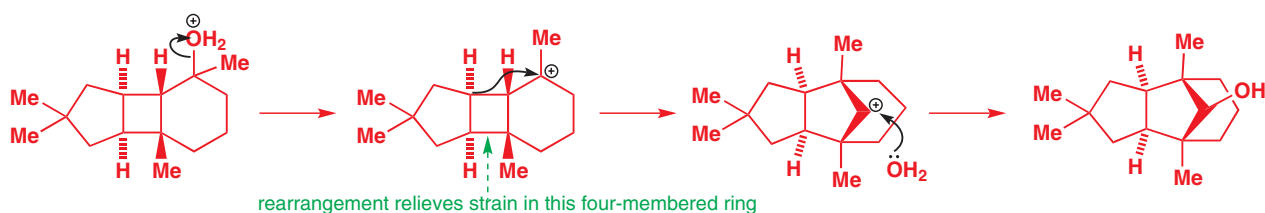
Although the rearrangement step transforms a stable tertiary cation into a less stable secondary cation, relief of strain in expansion from a four- to a five-membered ring makes the alkyl migration favourable. A synthesis of the natural product α -caryophyllene alcohol makes use of a similar ring expansion. Notice the photochemical [2 + 2] cycloaddition (Chapter 34) in the synthesis of the starting material.



Rearrangement of this tertiary alcohol in acid gives the target natural product. The four-membered ring has certainly disappeared but it may not be obvious at first what has taken its place.

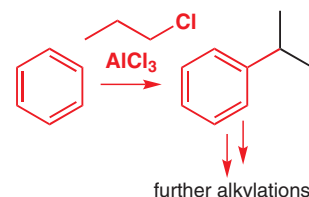


As usual, numbering the atoms makes clear what has happened: carbon 7 has migrated from carbon 6 to carbon 5. Loss of water gives a tertiary carbocation that undergoes rearrangement to a secondary carbocation with expansion of a four- to a five-membered ring.



Carbocation rearrangements: blessing or curse?

Well, that depends. You have now seen a few useful carbocation rearrangements that give single products in high yield. But you have also met at least one reaction that *cannot* be done because of carbocation rearrangements: as we mentioned in Chapter 15, Friedel–Crafts alkylation using primary alkyl halides leads to products derived from rearranged cations. The alkylation in the margin illustrates the problems of trying to use carbocation rearrangements to make single products in high yield. We can give three guidelines to spotting this type of reaction.

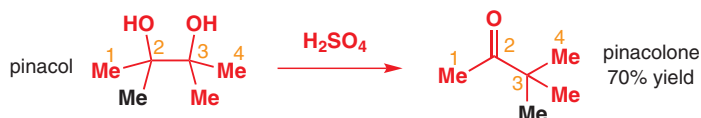


1. The rearrangement must be fast so that other reactions do not compete.
2. The product cation must be sufficiently more stable than the starting one so that the rearrangement happens in high yield.
3. Subsequent trapping of the product cation must be reliable: cations are high-energy intermediates, and are therefore unselective about how they react.

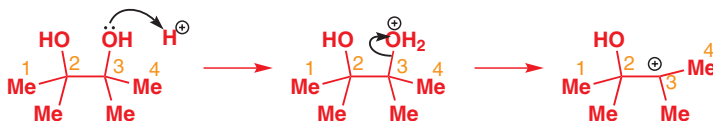
A reaction is no good if the cation reacts in more than one way—it may react with a nucleophile, eliminate, or undergo further rearrangement—but it must do only one of these! For the rest of the chapter, we will address only reactions that, unlike this Friedel–Crafts reaction, follow these guidelines. The reactions we will talk about all happen in good yield.

The pinacol rearrangement

When the 1,2-diol pinacol is treated with acid, a rearrangement takes place.



Whenever you see a rearrangement, especially in acid, you should now think ‘carbocation’. Here, protonation of one of the hydroxyl groups allows it to leave as water, giving the carbocation.

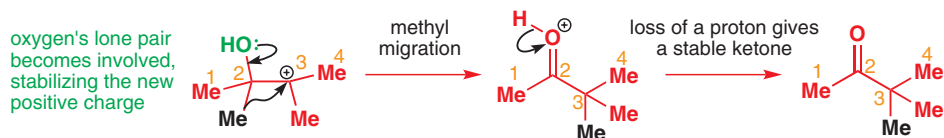


You now know that carbocations rearrange by alkyl shifts to get as stable as they can be—but this carbocation is already tertiary, and there is no ring strain, so why should it rearrange? Well, here we have another source of electrons to stabilize the carbocation: lone pairs on an oxygen atom. We pointed out early in the chapter that oxygen is very good at stabilizing a positive charge on an adjacent atom, and somewhat less good at stabilizing a positive charge two atoms away. By rearranging, the first-formed carbocation gets the positive charge into a position where the oxygen can stabilize it, and loss of a proton from oxygen then gives a stable ketone.

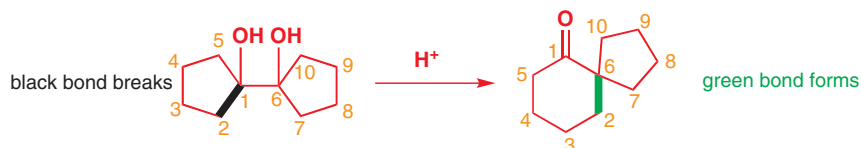
■ Pinacol, the trivial name for the starting material, which is made from acetone by a reaction you will meet in Chapter 37, gives its name to this class of rearrangements, and to the product, ‘pinacolone’.

Interactive mechanism for pinacol rearrangement

■ Spirocycles are pairs of rings joined at a single carbon atom (Chapter 32).

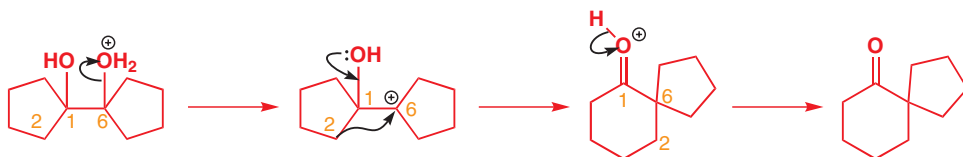


You can view the pinacol as a rearrangement with a 'push' and a 'pull'. The carbocation left by the departure of water 'pulls' the migrating group across at the same time as the oxygen's lone pair 'pushes' it. A particularly valuable type of pinacol rearrangement forms spirocyclic ring systems. You may find this one harder to follow, although the mechanism is identical with that of the last example. Our 'top tip' of numbering the atoms should help you to see what has happened: atom 2 has migrated from atom 1 to atom 6.



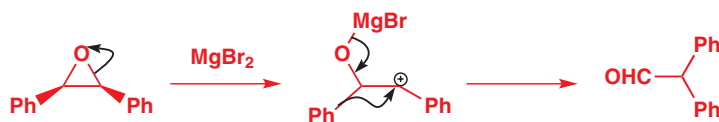
■ Of course, it doesn't matter how you number the atoms, but the numbering must be consistent. Usually, your initial impression of a greatly changed molecule will come down to just one or two atoms changing their substitution pattern, and numbering will help you to work out which ones they are.

When drawing the mechanism it doesn't matter which hydroxyl group you protonate or which adjacent C–C bond migrates—they are all the same. One five-membered ring expands to a six-membered ring but the reason this reaction happens is the formation of a carbonyl group, as in all pinacol rearrangements.

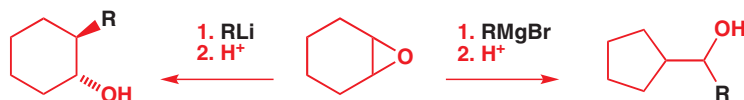


Epoxides rearrange with Lewis acids in a pinacol fashion

The intermediate cation in a pinacol rearrangement can equally well be formed from an epoxide, and treating epoxides with acid, including Lewis acids such as MgBr_2 , promotes the same type of reaction.



Rearrangement of epoxides with magnesium salts means that opening epoxides with Grignard reagents can give surprising results.



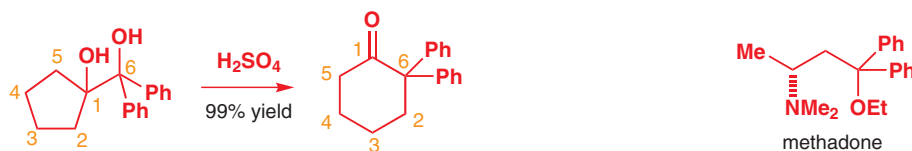
The alkyllithium reaction is quite straightforward as long as the alkyllithium is free of lithium salts. A clue to what has happened with the Grignard reagents comes from the fact that treating this epoxide with just MgBr_2 (not RMgBr) gives an aldehyde.



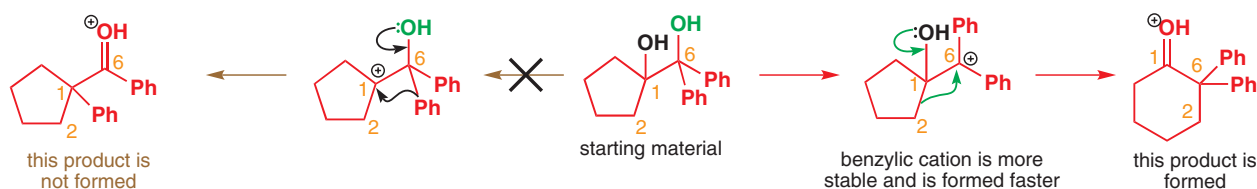
With a Grignard reagent, rearrangement occurs faster than addition to the epoxide, and then the Grignard reagent adds to the aldehyde.

Some pinacol rearrangements have a choice of migrating group

With these symmetrical diols and epoxides, it does not matter which hydroxyl group is protonated and leaves, nor which end the epoxide opens, nor which group migrates. When an unsymmetrical diol or epoxide rearranges, it is important which way the reaction goes. Usually, the reaction leaves behind the more stable cation. So, for example, this unsymmetrical diol gives the ring-expanded ketone, a starting material for the synthesis of analogues of the drug methadone.



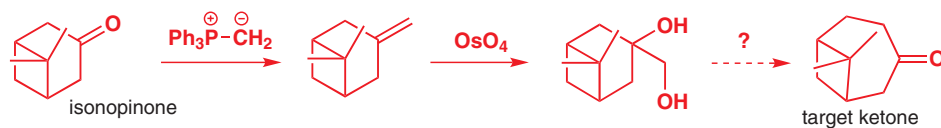
This product is formed because the green OH group leaves more readily than the black as the carbocation stabilized by two phenyl groups forms more readily than the carbocation stabilized by two alkyl groups. The migration step which follows has no choice: both alkyl groups on the black alcohol are the same.



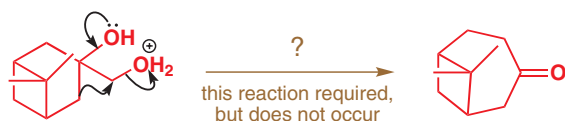
Most unsymmetrical diols or epoxides give mixtures of products on rearrangement. The problem is that there is a choice of two leaving groups and two alternative rearrangement directions, and only for certain substitution patterns is the choice clear-cut.

Semipinacol rearrangements are pinacol reactions with no choice about which way to go

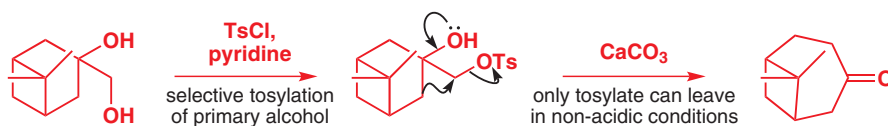
For some work on perfumery compounds, this seven-membered cyclic ketone was needed. A reasonable starting material to use is the diol shown because it can be made in two steps from the natural product isonopinone.



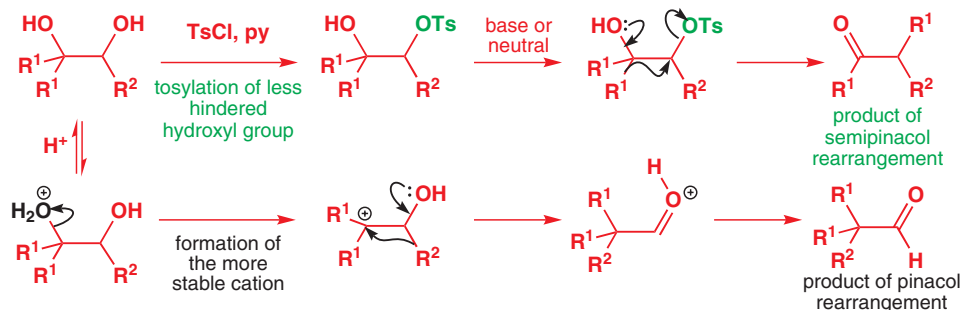
The reaction needed for the last stage is a pinacol rearrangement—the *primary* hydroxyl group needs persuading to leave as the ring expands. The problem is, of course, that the tertiary hydroxyl group is much more likely to leave since it leaves behind a more stable carbocation.



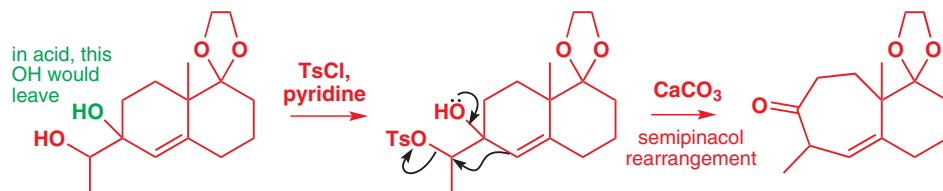
The solution to this problem is to force the primary hydroxyl group to be the leaving group by making it into a tosylate. The primary hydroxyl group reacts more rapidly with TsCl than the tertiary one because it is less hindered. A weak base is now all that is needed to make the compound rearrange in what is known as a semipinacol rearrangement.



Semipinacol rearrangements are rearrangements in which a hydroxyl group provides the electrons to 'push' the migrating group across, but the 'pull' comes from the departure of leaving groups other than water—tosylate in this example, but typically also halide or nitrogen (N_2). Since tosylation occurs at the *less* hindered hydroxyl group of a diol, not only can semipinacol rearrangements be more regioselective than pinacol rearrangements, but their regioselectivity may be in the opposite direction.

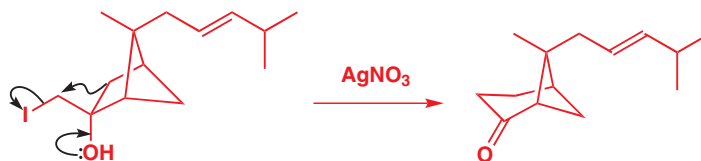


Corey exploited this in a synthesis of the natural product longifolene. He needed to persuade an easily made 6,6-fused ring system to undergo rearrangement to a ring-expanded ketone. Again, a normal acid-catalysed pinacol rearrangement is no good—the tertiary, allylic hydroxyl group is much more likely to ionize, and the acid-sensitive protecting group would be hydrolysed too. Tosylation of the secondary alcohol in the presence of the tertiary is possible, and semipinacol rearrangement gives the required ketone.



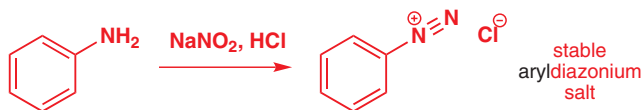
■ Treating 2-halo alcohols with base is, of course, a good way to make epoxides. Using AgNO_3 to improve iodide's leaving ability without increasing the nucleophilicity of the hydroxyl group favours rearrangement at the expense of epoxide formation. There would certainly be a danger of epoxide formation in strong base.

The leaving group need not be tosylate: in the following example, part of a synthesis of bergamotene (a component of valerian root oil and the aroma of Earl Grey tea), a 2-iodo alcohol rearranges.



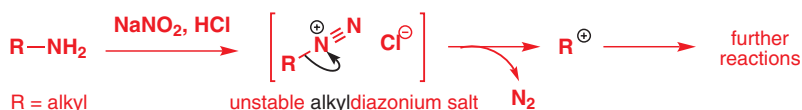
Semipinacol rearrangements of diazonium salts

You saw in Chapter 21 how aromatic amines can be converted to diazonium salts by treatment with acidic sodium nitrite.

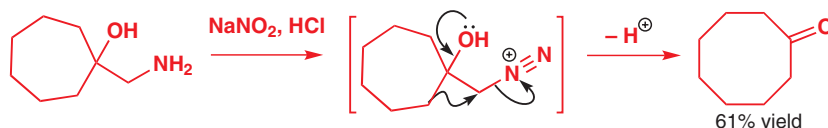


■ It might be an idea to review pp. 520–523 of Chapter 22 to be sure you understand the mechanism of this reaction.

Aryldiazonium salts are stable but alkyldiazonium salts are not: nitrogen gas is the world's best leaving group, and, when it goes it leaves behind a carbocation.



One of the 'further reactions' this carbocation can undergo is rearrangement. If the starting amine is a 2-amino alcohol, the cation can be stabilized by a semipinacol rearrangement.

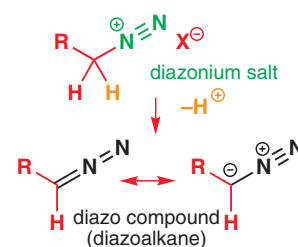
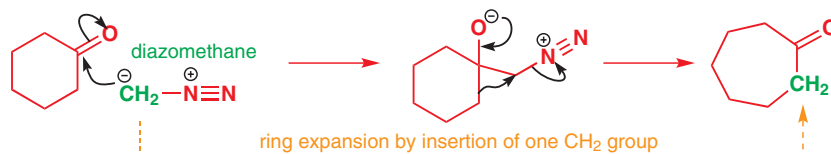


■ Semipinacol rearrangements of diazonium salts derived from 2-amino alcohols are sometimes called **Tiffeneau–Demjanov rearrangements**.

Interactive mechanism for Tiffeneau–Demjanov rearrangement

While alkyldiazonium salts are unstable, their conjugate bases, diazoalkanes, are stable enough to be prepared and are nucleophilic towards carbonyl compounds. Diazoalkanes are neutral compounds having one fewer proton than diazonium salts, and are delocalized structures with a central sp nitrogen atom.

When diazomethane (a compound we will investigate in more detail in Chapter 38) adds to a ketone, the product undergoes a ring expansion by rearrangement of the same type of intermediate.

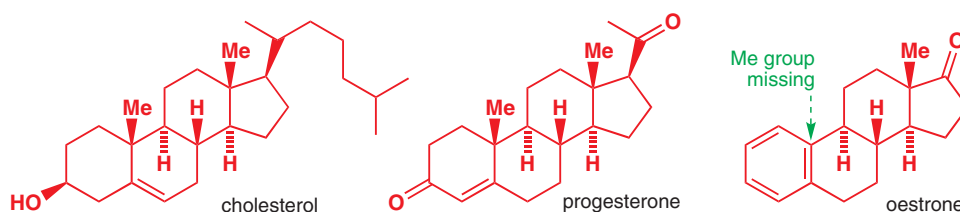


The problem with reactions like this is that both the starting material and product are ketones, so they work cleanly only if the starting material is more reactive than the product. Cyclohexanone is more reactive as an electrophile than either cyclopentanone or cycloheptanone, so it ring expands cleanly to cycloheptanone. But expansion of cyclopentanone to cyclohexanone is messy and gives a mixture of products.

Interactive mechanism for semipinacol rearrangements of diazonium salts

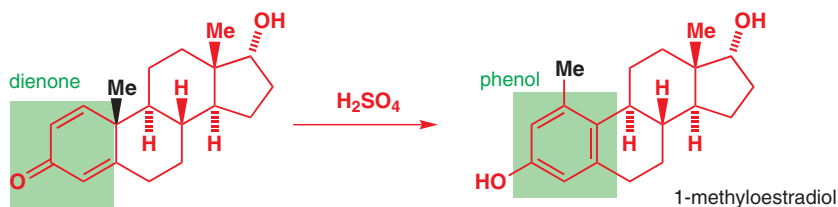
The dienone-phenol rearrangement

The female sex hormone oestrone is the metabolic product of another hormone, progesterone, itself made in the body from cholesterol.

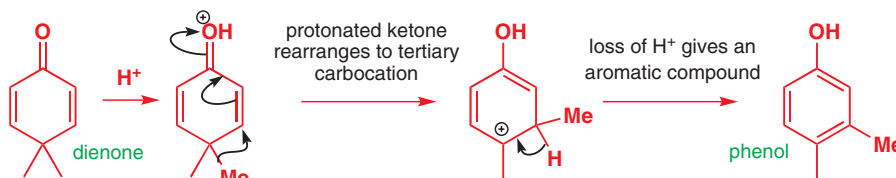


Oestrone lacks one of progesterone's methyl groups, probably removed in the body as CO₂ after oxidation. In 1946, Carl Djerassi, a man whose work led directly to the invention of the contraceptive pill, showed that another derivative of cholesterol could be rearranged to the oestrone analogue 1-methyloestradiol—notice how the methyl group has this time migrated to an adjacent carbon atom. At the same time, the dienone has become a phenol.

Carl Djerassi, an American born in Vienna in 1923, worked chiefly at Ciba, at Syntex in Mexico, and at Stanford. He developed syntheses of human steroids from compounds in plants, was a pioneer of mass spectrometry, and is a colourful campaigner for peace and disarmament, along with being a playwright and novelist.

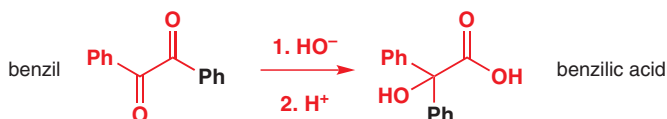


This type of rearrangement is known helpfully as a **dienone-phenol rearrangement**, and we can consider it quite simply as a type of *reverse* pinacol rearrangement. Pinacol and semipinacol rearrangements are driven by the formation of a carbonyl group. The rearranged cation is stabilized by being next to oxygen and it can rapidly lose H^+ to give a carbonyl compound. In the key step of a dienone-phenol rearrangement, a protonated carbonyl compound rearranges to a tertiary carbocation. The reaction is driven from dienone to phenol because the product cation can rapidly undergo elimination of H^+ to become aromatic.

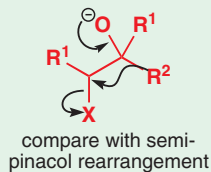


The benzilic acid rearrangement

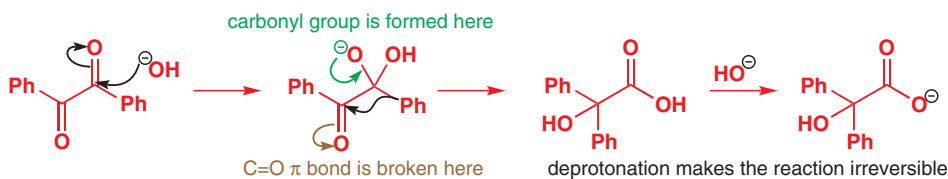
You have seen rearrangements in which carbonyl groups form at the migration origin: the migrating group in the pinacol and semipinacol rearrangements is 'pushed' by the oxygen's lone pair as it forms the new carbonyl group. You have also seen carbonyl groups being destroyed at the migration terminus: the migrating group in the dienone-phenol rearrangement is 'pulled' towards the protonated carbonyl group. The first rearrangement reaction ever to be described has both of these at once.



■ You may find it helpful to think of the benzilic acid rearrangement as a semipinacol rearrangement in which we have a breaking $\text{C}=\text{O}$ π bond instead of a leaving group.



In 1838, Justus von Liebig found that treating 'benzil' (1,2-diphenylethane-1,2-dione) with hydroxide gave, after acid quench, 2-hydroxy-2,2-diphenylacetic acid, which he called 'benzilic acid'. The mechanism of this benzilic acid rearrangement starts with attack of hydroxide on one of the carbonyl groups. The tetrahedral intermediate can collapse in a reaction reminiscent of a semipinacol rearrangement.



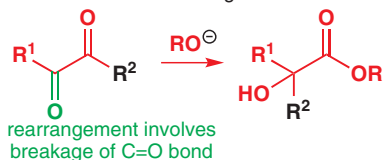
Interactive mechanism for benzilic acid rearrangement

The Favorskii rearrangement

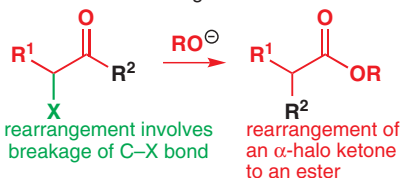
We hope you have appreciated the smooth mechanistic progression so far in this chapter, from Wagner-Meerwein to pinacol and semipinacol through dienone-phenol to benzilic acid.

Our aim is to help you gain an overall view of the types of rearrangements that take place (and why) and not to present you with lots of disconnected facts. It is at this point, however, that our mechanistic journey takes a hairpin bend. The bend comes as a surprise because when we show you the next rearrangement, the Favorskii, you would be forgiven for thinking that surely it's just a variant of the benzilic acid rearrangement?

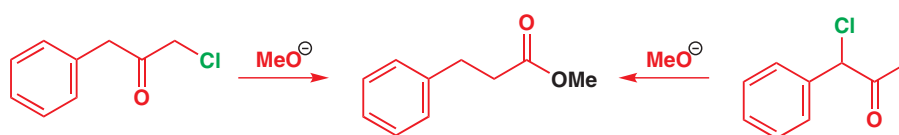
the benzilic acid rearrangement



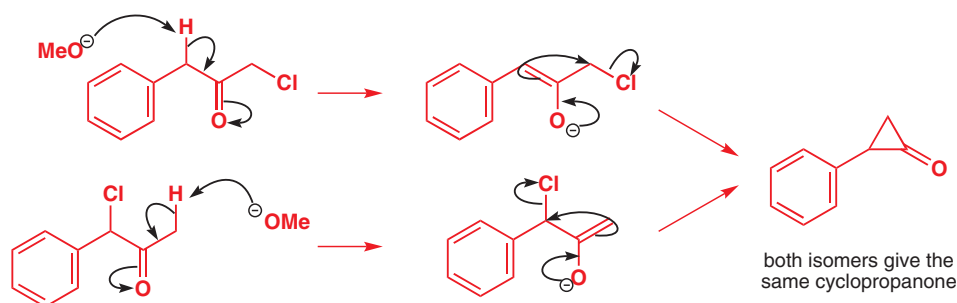
the Favorskii rearrangement



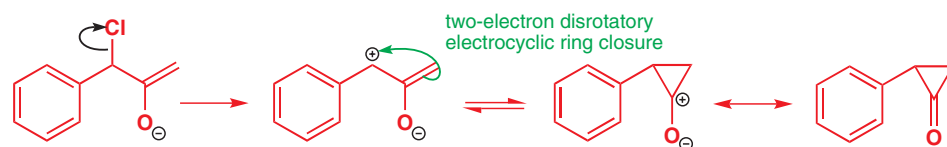
Well, this is what chemists thought until 1944, when some American chemists found that two isomeric α -chloro ketones gave exactly the same product on treatment with methoxide. They suggested that both reactions went through the same intermediate.



That intermediate is a three-membered cyclic ketone, a cyclopropanone: the alkoxide acts not as a nucleophile (its role in the benzilic acid rearrangement) but as a base, enolizing the ketone. The enolate can alkylate itself intramolecularly in a reaction that looks bizarre but that many chemists think is not unreasonable. The intermediate is the same cyclopropanone in each case.

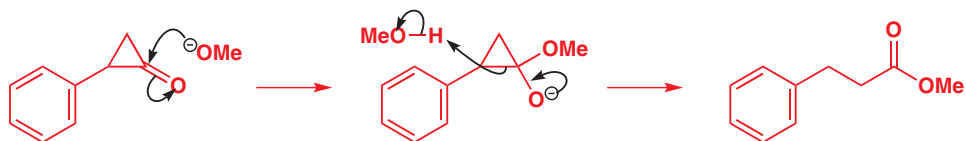


There is also a pericyclic mechanism for the ring-closure step. The enolate simply loses chloride to give an 'oxyallyl cation'—a dipolar species with an oxyanion and a delocalized allylic cation. This species can cyclize in a two-electron disrotatory electrocyclic reaction (Chapter 35) to give the same cyclopropanone.

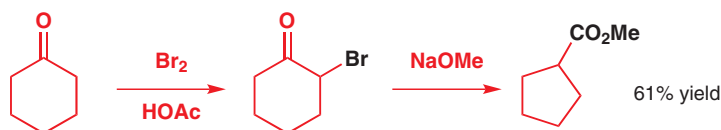


Cyclopropanones are very reactive towards nucleophiles, and the tetrahedral intermediate arising from the attack of methoxide springs open to give the ester product. The more stable carbanion leaves: although the carbanion is not actually formed as a free species, there must be considerable negative charge at the carbon atom as the three-membered ring opens. Here the benzyl group is the better leaving group.

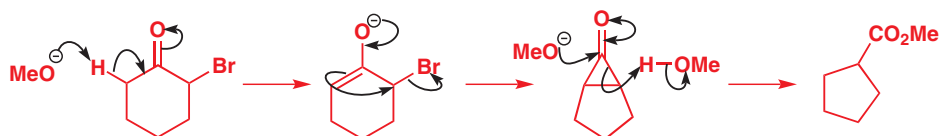
Cyclopropanones and cyclobutanones are very reactive, rather like epoxides, because, while the 60° or 90° angle in the ring is nowhere near the tetrahedral angle (108°), it is nearer 108° than the 120° preferred by the sp^2 C of the C=O group. Conversely, the small ring ketones are resistant to enolization because that would place two sp^2 carbon atoms in the ring.



Favorskii rearrangement of cyclic 2-bromoketones leads to ring contraction and this has become one of the most fruitful uses of the rearrangement in synthesis. Bromination of cyclohexanone (Chapter 20) and treatment with methoxide gives the methyl ester of cyclopentane carboxylic acid in good yield.



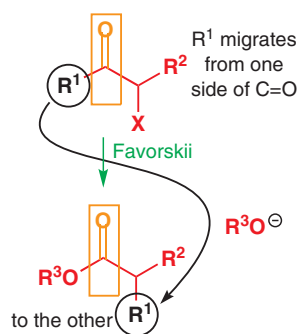
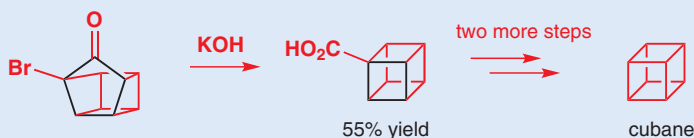
Enolization occurs on the side of the ketone away from the bromine atom and the enolate cyclizes as before but the cyclopropanone intermediate is symmetrical so that the product is the same whichever C–C bond breaks after nucleophilic attack by the methoxide ion.



Interactive mechanism for the Favorskii rearrangement

Cubane from a Favorskii rearrangement

In 1964, two American chemists synthesized for the first time a remarkable molecule, cubane. Two of the key steps were Favorskii rearrangements, which allowed the chemists to contract five-membered rings to four-membered rings. Here is one of them. Two more steps decarboxylate the product to give cubane itself.

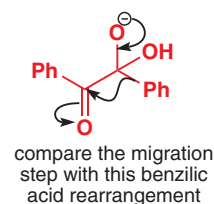
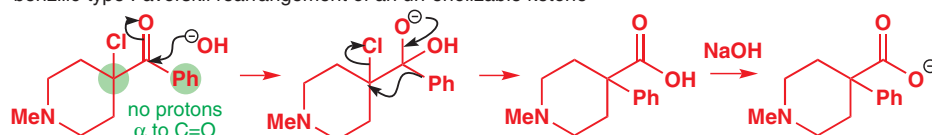


The overall consequence of the Favorskii rearrangement is that an alkyl group is transferred from one side of a carbonyl group to the other. This means that it can be used to build up heavily branched esters and carboxylic acids—the sort that are hard to make by alkylation (Chapter 25) because of the problems of hindered enolates and unreactive secondary alkyl halides. Heavily substituted acids, where CO_2H is attached to a tertiary carbon atom, would be hard to make by any other method.



The Favorskii rearrangement is also a key step in the synthesis of the powerful obstetric painkiller pethidine. But try writing a mechanism for this last reaction and you run into a problem—there are no acidic protons so the ketone cannot be enolized! Yet the Favorskii rearrangement still works. Despite our warnings against confusing the mechanisms of the Favorskii and benzilic acid rearrangements, the Favorskii rearrangement may, in fact, follow a benzilic-type rearrangement mechanism, *if* there are no acidic hydrogens available.

benzilic-type Favorskii rearrangement of an un-enolizable ketone

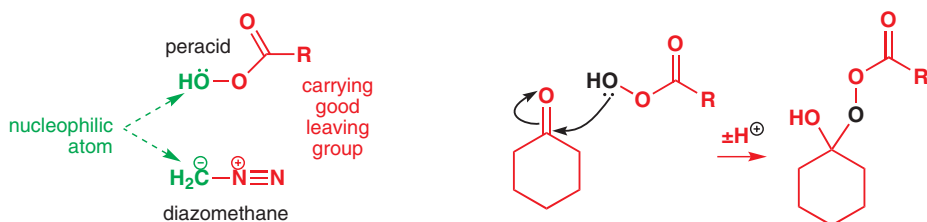


Migration to oxygen: the Baeyer–Villiger reaction

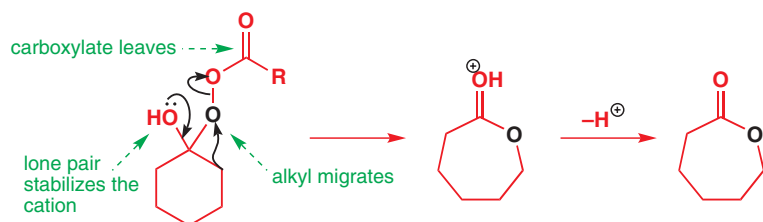
In 1899, two Germans, A. Baeyer and V. Villiger, found that treating a ketone with a peracid (RCO_3H) can produce an ester. An oxygen atom is ‘inserted’ next to the carbonyl group. You saw a similar ‘insertion’ reaction earlier in the chapter, and the mechanism here is not dissimilar.



Both peracids and diazomethane contain a nucleophilic centre that carries a good leaving group, and addition of peracid to the carbonyl group gives a structure that should remind you of a semipinacol intermediate with one of the carbon atoms replaced by oxygen.



Carboxylates are not such good leaving groups as nitrogen, but the oxygen–oxygen single bond is very weak. Once the peracid has added, loss of carboxylate is concerted with a rearrangement driven by formation of a carbonyl group.

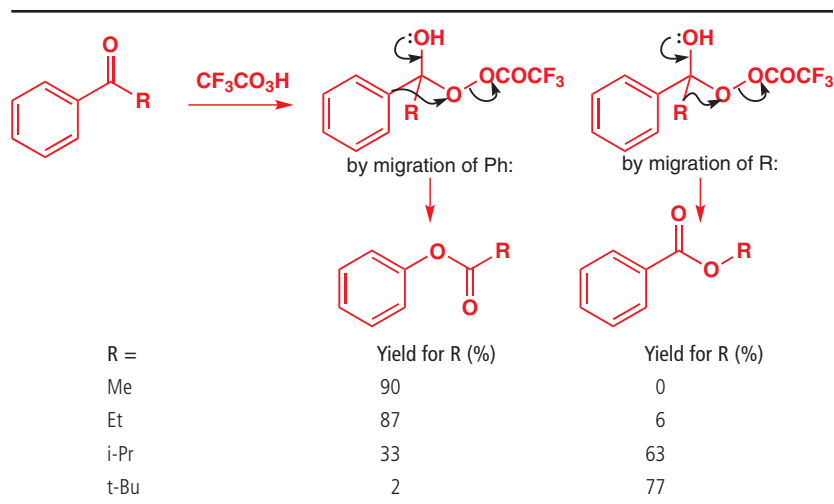


Baeyer–Villiger reactions are among the most useful of all rearrangement reactions, and the most common reagent is *m*-CPBA (*meta*-chloroperbenzoic acid) because it is commercially available.

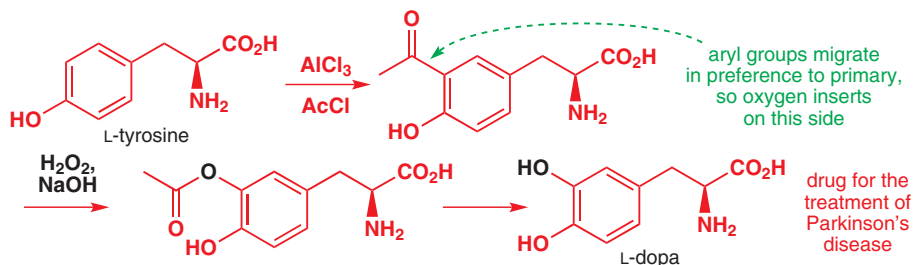
Which group migrates? (I): the facts

A question we have deliberately avoided up to this point is this: when there is a competition between two migrating groups, *which group migrates*? This question arises in pinacol, semipinacol, and dienone-phenol rearrangements and in Baeyer–Villiger reactions (in the benzilic acid and Favorskii rearrangements, there is no choice) and the awkward fact is that the answer is different in each case! However, let's start with the Baeyer–Villiger reaction because here the question is always valid except when the ketone being oxidized is symmetrical. Here are some examples; you can probably begin to draw up guidelines for yourself.

 Interactive mechanism for the Baeyer–Villiger rearrangement



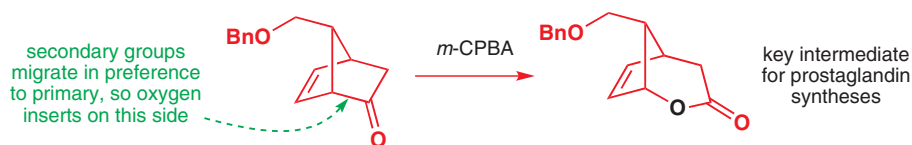
The order, with *tert*-alkyl the best at migrating, then *sec*-alkyl closely followed by Ph, then Et, then Me, *very roughly* follows the order in which the groups are able to stabilize a positive charge. Primary groups are much more reluctant to undergo migration than secondary ones or aryl groups, and this makes regioselective Baeyer–Villiger reactions possible.



The Baeyer–Villiger reaction has solved a regioselectivity problem here. L-tyrosine, a relatively cheap amino acid, can be converted to the important drug L-dopa provided it can be hydroxylated *ortho* to the OH group. This is where electrophilic substitutions of the phenol take place, but electrophilic substitutions with ‘HO⁺’ are not possible. However, after a Friedel–Crafts acylation, the acyl group can be converted to hydroxyl by the Baeyer–Villiger reaction and hydrolysis. The Baeyer–Villiger reaction means that MeCO⁺ can be used as a synthetic equivalent for ‘HO⁺’. Note the unusual use of the less reactive H₂O₂ as oxidizing agent in this reaction. This is possible only when the migrating group is an electron-rich aromatic ring; these reactions are sometimes called Dakin reactions.

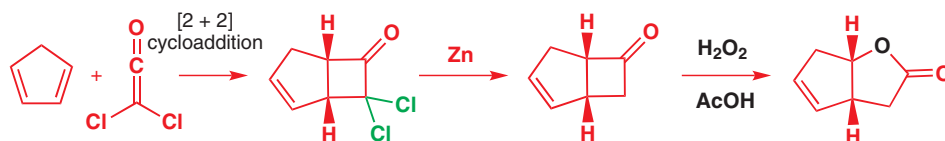
Unsaturated ketones may epoxidize or undergo Baeyer–Villiger rearrangement

Peracids may epoxidize alkenes faster than ketones take part in Baeyer–Villiger reactions, so unsaturated ketones are not often good substrates for Baeyer–Villiger reactions. The balance is rather delicate. The two factors that matter are: how *electrophilic* is the ketone and how *nucleophilic* is the alkene? You might like to consider why this reaction *does* work, and why the C=C double bond here is particularly unreactive.

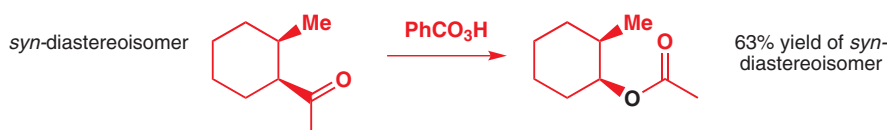


Small-ring ketones can relieve ring strain by undergoing Baeyer–Villiger reactions—this cyclobutanone (an intermediate in a synthesis of the perfumery compound *cis*-jasmone) is

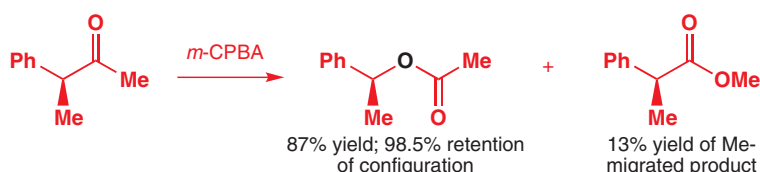
made by a ketene [2 + 2] cycloaddition, and is so reactive that it needs only H_2O_2 to rearrange. Unlike $\text{CF}_3\text{CO}_3\text{H}$ or *m*-CPBA, H_2O_2 will not epoxidize double bonds unless they are electron-deficient (see Chapter 22).



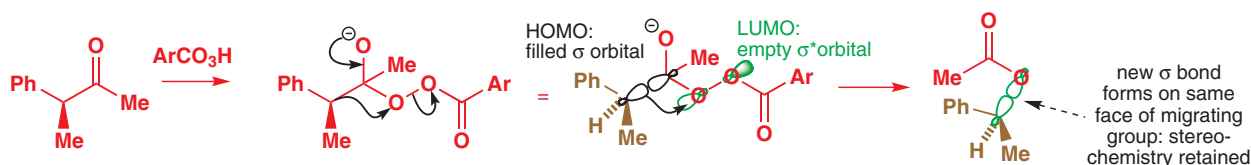
One point to note about both of the last two reactions is that the insertion of oxygen goes with retention of stereochemistry. You may think this is unsurprising in a cyclic system like this and, indeed, the first of the two cannot possibly go with inversion. However, this is a general feature of Baeyer–Villiger reactions, even when inversion would give a more stable product.



Even when you might imagine that racemization would occur, as in this benzylic ketone, retention is the rule.



By looking at the orbitals involved, you can see why this must be so. The sp^3 orbital of the migrating carbon just slips from one orbital to the next with the minimum amount of structural reorganization. The large lobe of the sp^3 orbital is used so the new bond forms to the same face of the migrating group as the old one, and stereochemistry is retained.

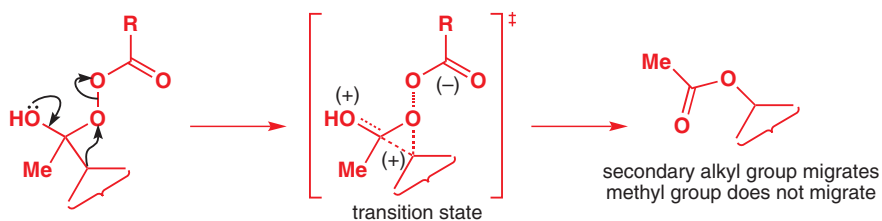


The orbital interactions in all 1,2-migrations are similar, and the migrating group retains its stereochemistry in these too. In the more familiar $\text{S}_{\text{N}}2$ reaction, inversion occurs because the antibonding σ^* orbital rather than the bonding σ orbital is used. In the $\text{S}_{\text{N}}2$ reaction, carbon undergoes *nucleophilic* attack with *inversion*; in rearrangements the migrating carbon atom undergoes *electrophilic* attack with *retention* of configuration.

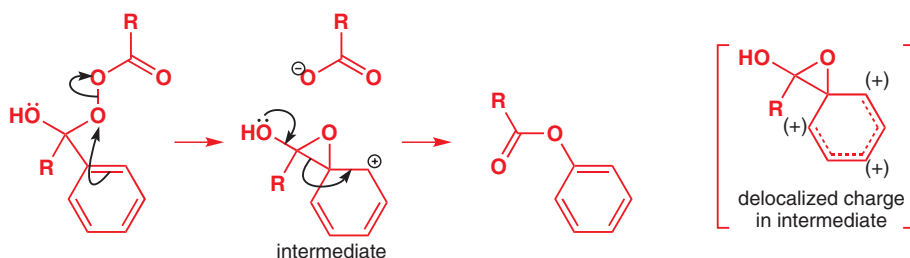
- In 1,2-migrations, the migrating group retains its stereochemistry.

Which group migrates? (II): the reasons

Why does the more substituted group migrate in the Baeyer–Villiger reaction? The transition state has a positive charge spread out over the molecule as the carboxylate leaves as an anion. If the migrating group can take some responsibility for the positive charge the transition state will be more stable. The more stable the charge, the faster the rearrangement.



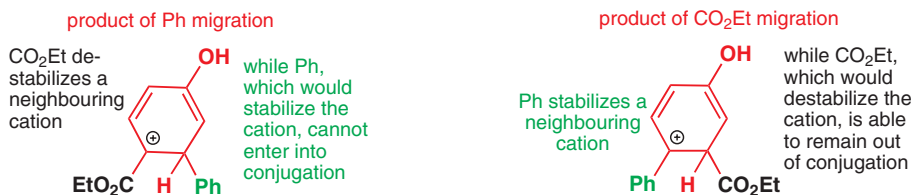
When a benzene ring migrates, π participation is involved as the benzene ring acts as a nucleophile and the positive charge can be spread out even further. Note that the Ph is stabilizing the charge here in the way that it stabilizes the intermediate in an electrophilic aromatic substitution reaction—like a pentadienyl cation rather than like a benzylic cation. What was a transition state in alkyl migration becomes an intermediate in phenyl migration.



The situation in other rearrangements is much more complicated—and indeed more complicated than many textbooks would have you believe. We shall look just briefly at the dienone-phenol rearrangement again, this time considering reactions in which there is competition between two different migrating groups. As in the Baeyer–Villiger reaction, the transition state is cationic, so you would expect cation-stabilizing groups to migrate more readily. This appears to be true for Ph versus Me, but is most definitely not true for Ph versus CO_2Et . The cation *destabilizing* group CO_2Et migrates even though Ph is much better at stabilizing a positive charge!



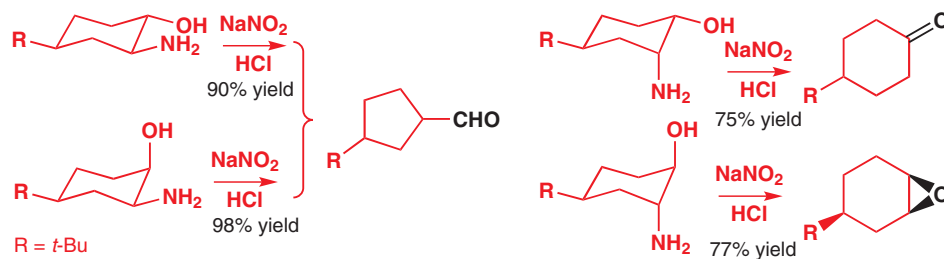
The reason is that CO_2Et is so cation *destabilizing* that it prefers to migrate rather than be left behind next door to a cation. In this case, then, it is the cation-stabilizing ability of the group that *does not* migrate that matters most.



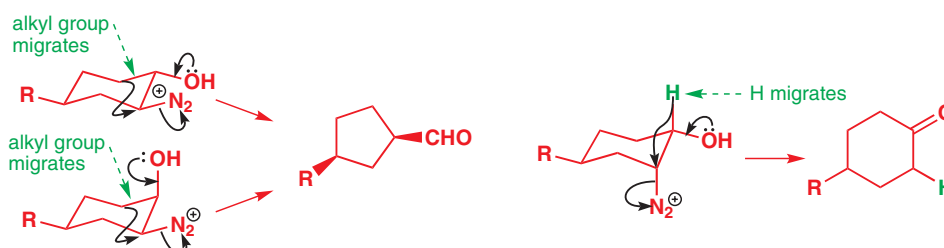
Which group migrates? (III): stereochemistry matters too

Selectivity in rearrangement reactions is affected by the electronic nature of *both* the group that migrates *and* the group that is left behind. But there is more! *Stereochemistry* is important too. The outcome of diazotization and semipinacol rearrangement (Tiffeneau–Demjanov

rearrangement, p. 949) of this amino-alcohol depends entirely on the diastereoisomer you start with. There are four diastereoisomers, and we have drawn each one in the only conformation it can reasonably adopt, with the *t*-butyl group equatorial.

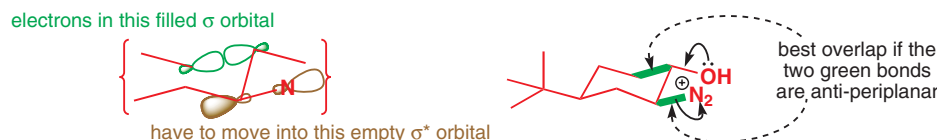


In all of these reactions, the OH group provides the electronic ‘push’. In the first two reactions, the ring contracts by an alkyl migration from the secondary alcohol, while in the third it is H that migrates from the same position.

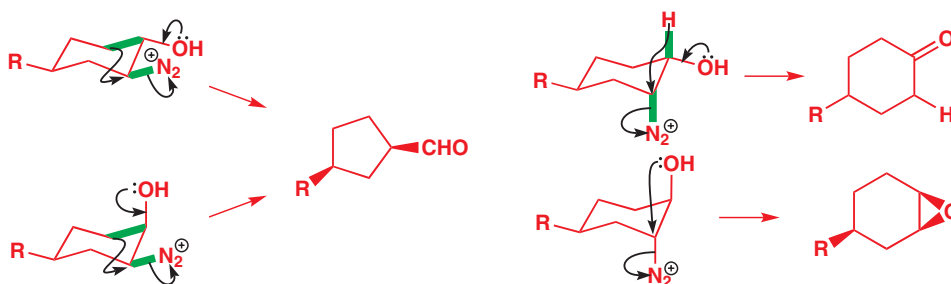


The only difference between the compounds is stereochemistry and, if we look at the orbitals involved in the reactions, we can see why this is so important. As the N_2 leaving group departs, electrons in the bond to the migrating group have to flow into the C–N σ^* orbital—we discussed this on p. 949. But what we didn’t talk about then was the fact that best overlap between these two orbitals (σ and σ^*) occurs if they are anti-periplanar to one another—just as in an E2 elimination reaction.

 Interactive explanation of the stereochemistry of rearrangements



For the first two compounds, with the $-\text{N}_2^+$ group equatorial, the group best placed to migrate is the alkyl group that forms the ring; for the third reaction, there is a hydrogen atom anti-periplanar to the leaving group, so H migrates.

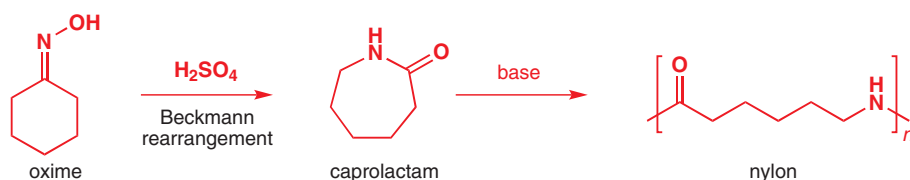


The fourth reaction has, rather than a group that might migrate, the hydroxyl group ideally placed to displace N_2 and form an epoxide.

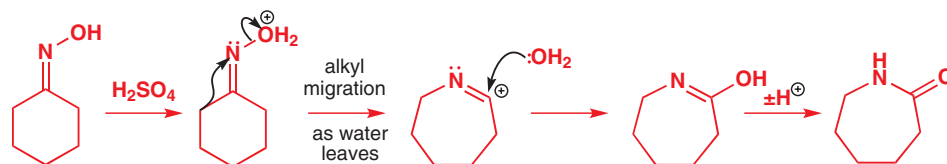
The requirement for the migrating group to be anti-periplanar to the leaving group is quite general in rearrangement reactions. The reason we haven't noticed its effect before is that most of the compounds we have considered have not been conformationally constrained in the way that these are. Free rotation means that the right geometry for rearrangement is always obtainable—stereochemistry is not a factor in the Baeyer–Villiger reaction, for example. We will come back to some more aspects of stereochemical control later in the chapter, when we deal with fragmentation reactions. Before then, we will consider one last rearrangement reaction, in which stereochemistry again plays an important controlling role.

The Beckmann rearrangement

The industrial manufacture of nylon relies on the alkaline polymerization of a cyclic amide known trivially as caprolactam. Caprolactam can be produced by the action of sulfuric acid on the oxime of cyclohexanone in a rearrangement known as the **Beckmann rearrangement**.

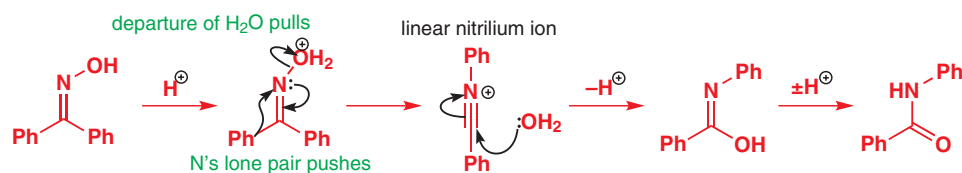


The mechanism of the Beckman rearrangement follows the same pattern as a pinacol or Baeyer–Villiger reaction: acid converts the oxime OH into a leaving group, and an alkyl group migrates to nitrogen as water departs. The product cation is then trapped by water to give an amide.



Interactive mechanism for the Beckmann rearrangement

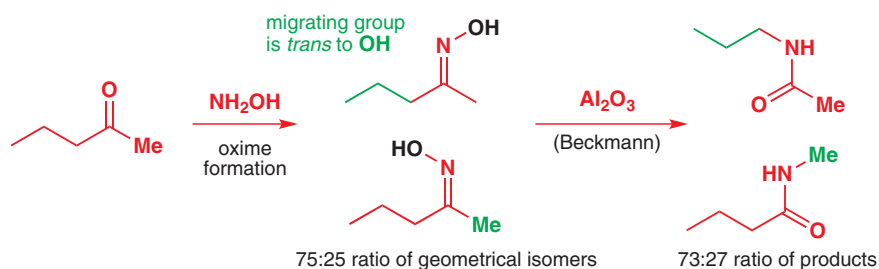
This rearrangement is not confined to cyclic oximes, and other ways of converting OH to a leaving group also work, such as PCl_5 , SOCl_2 , and other acyl or sulfonyl chlorides. In an acyclic Beckmann rearrangement, the product cation is better represented as this nitrilium ion. When we write the mechanism we can then involve the nitrogen's lone pair to 'push' the migrating group back on to N.



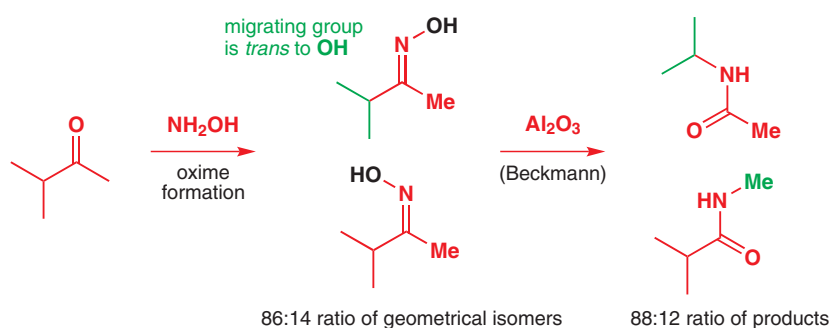
■ A linear nitrilium ion like this is impossible in a seven-membered ring of the last example.

Which group migrates in the Beckmann rearrangement?

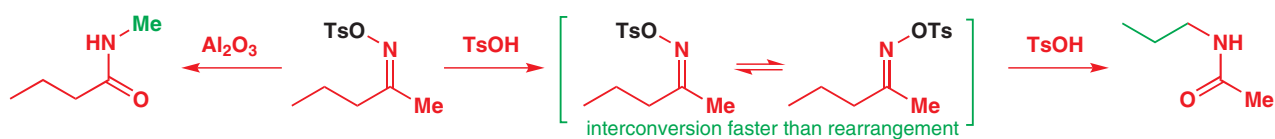
In the Beckmann rearrangement of unsymmetrical ketones there are two groups that could migrate. There are also two possible geometrical isomers of an unsymmetrical oxime: $\text{C}=\text{N}$ double bonds can exhibit *cis/trans* isomerism just as $\text{C}=\text{C}$ double bonds can. When mixtures of geometrical isomers of oximes are rearranged, mixtures of products result, but the ratio of products mirrors exactly the ratio of geometrical isomers in the starting materials—the group that has migrated is in each case the group *trans* to the OH in the starting material.



We have already touched on the idea that, for migration to occur, a migrating group has to be able to interact with the σ^* of the bond to the leaving group, and this is the reason for the specificity here. In the example a couple of pages back the stereospecificity of the reaction was due to the starting material being constrained in a conformationally rigid ring. Here it is the C=N double bond that provides the constraint. If one of the alkyl chains is branched, more of the oxime with the OH group *anti* to that chain will be formed and correspondingly more of the branched group will migrate.



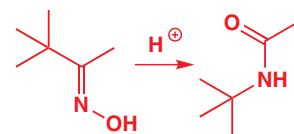
Conditions that allow those double bond isomers to interconvert can allow either group to migrate—which one does so will then be decided, as in the Baeyer–Villiger reaction, by electronic factors. Most protic acids allow the oxime isomers to equilibrate, so, for example, this tosylated oxime rearranges with full stereospecificity in Al_2O_3 (the *anti* methyl group migrates), but with TsOH , equilibration of the oxime geometrical isomers means that either group could migrate—in the event, the propyl group (which is more able to support a positive charge) migrates faster.

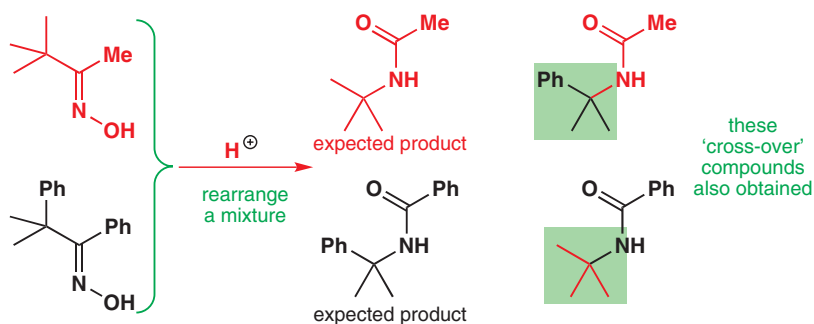


Notice that the effect of the Beckmann rearrangement is to insert a *nitrogen* atom next to the carbonyl group. It forms a useful trio with the Baeyer–Villiger *oxygen* insertion and the diazoalkane *carbon* insertion.

The Beckmann fragmentation

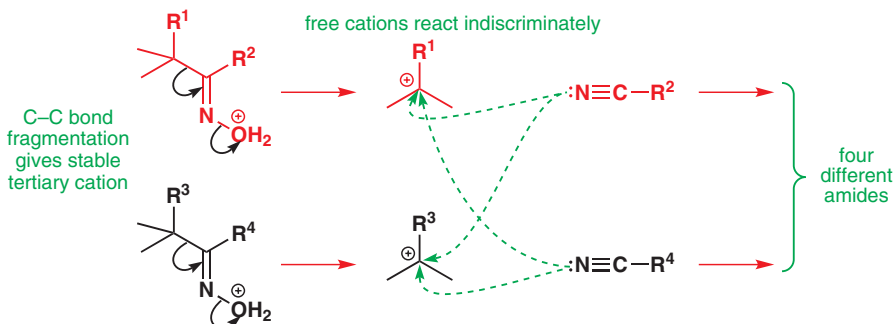
To introduce the theme of the last section of this chapter, a Beckmann rearrangement that is not all that it seems. *t*-Butyl groups migrate well in the Baeyer–Villiger reaction and, indeed, Beckmann rearrangement of the compound in the margin appears to be quite normal too. But, when this compound and another compound with a tertiary centre next to the oxime are mixed together and treated with acid, it becomes apparent that what is happening is not an intramolecular reaction.





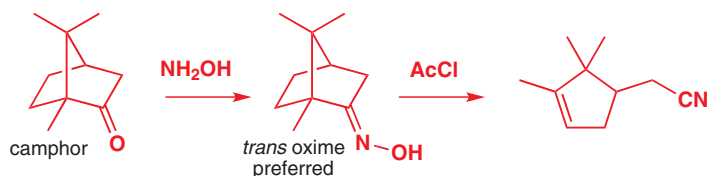
Each migrating tertiary group must have lost contact with the amide fragment it started out with. The molecule must fall apart to give a *t*-alkyl cation and a nitrile: the Beckmann rearrangement now goes via a fragmentation mechanism.

■ The recombination step of this reaction is really just a Ritter reaction: reaction of a nitrile with a carbocation. You came across the Ritter reaction on p. 353.



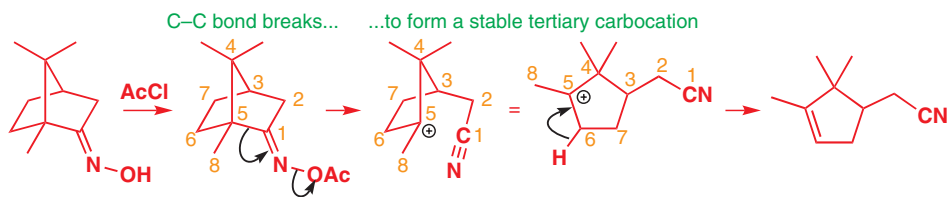
Migrating groups have to provide some degree of cation stabilization. But if they stabilize a cation *too well* there is a good chance that fragmentation will occur and the 'migrating group' will be lost as a carbocation.

Here is a more convincing example of the same fragmentation reaction: the conditions, but not the results, are those of a Beckmann rearrangement. In this reaction, the ring structure means the cation cannot be trapped by the nitrile, and a fragmentation product is isolated.



Beckmann rearrangements that go with fragmentation are sometimes called 'anomalous' or 'second-order' Beckmann rearrangements. You should not use the second of these names and, in any case, **Beckmann fragmentation** is much better than either.

The mechanism is straightforward once you know what happens to Beckmann rearrangements when the migrating group is tertiary—but hard to follow unless you number the atoms!



Interactive mechanism for Beckmann fragmentation

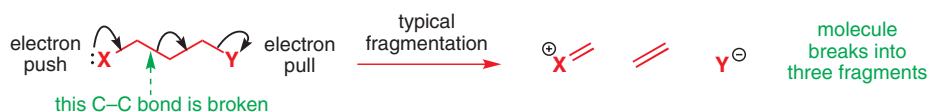
Polarization of C-C bonds helps fragmentation

Up to now, you have met few fragmentation reactions—reactions in which C-C bonds are broken—largely because the C-C bond is so strong. Why then does the Beckmann

fragmentation work? Well, the reason C–C bonds are hard to break is not just because of their strength, as the table of bond energies indicates.

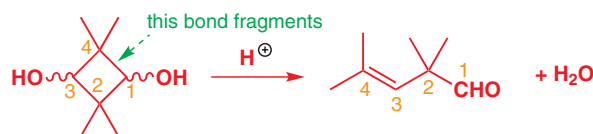
For both carbon and hydrogen, a bond to oxygen is *stronger* than a bond to carbon. Yet we have no hesitation in breaking O–H bonds (of, say, carboxylic acids) with even the weakest of bases and we have spent much of this chapter showing C–O bonds of protonated alcohols rupturing spontaneously! What is going on?

The answer is **polarization**. Oxygen's electronegativity means that C–O and O–H bonds are polarized and are easy to break with hard nucleophiles and bases; C–C and C–H bonds are (usually) not polarized and, although weaker, are harder to break. It follows that to break a C–C bond it helps a lot if it is polarized—there needs to be a source of electrons at one end and an electron 'sink' (into which they can flow) at the other.

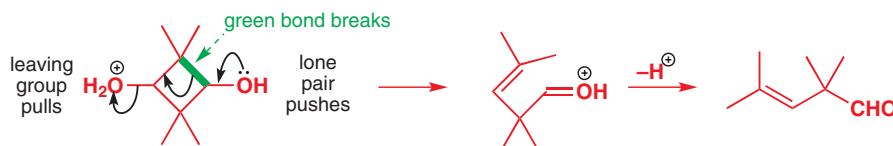


Fragmentations require electron push and electron pull

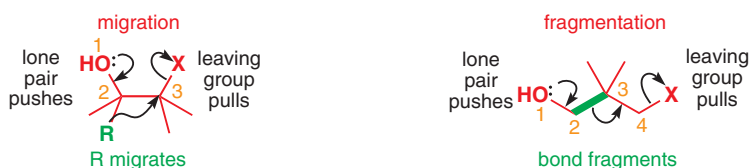
Fragmentations are reactions in which the molecule breaks open by the cleavage of a C–C single bond, and we start this section with some examples. Both diastereoisomers of this cyclic diol fragment in acid to give an aldehyde.



Numbering the atoms shows which bond fragments—now we need to provide a source and a sink for the electrons to polarize the bond. Protonation of a hydroxyl group provides the sink—it can now leave as water. And the lone pair of the other oxygen provides the source. You can think of the electrons in the C–C bond being 'pushed' by the oxygen's lone pair and 'pulled' by the departing water—until the bond breaks. A bit of extra impetus comes from release of ring strain: C–C bonds in three- and four-membered rings are weaker than usual (by about 120 kJ mol⁻¹).



We talked about 'pushing' and 'pulling' electrons when we introduced the pinacol rearrangement, and a very similar thing is happening here *but* the electron source and sink are *separated by one atom* instead of being *adjacent*.



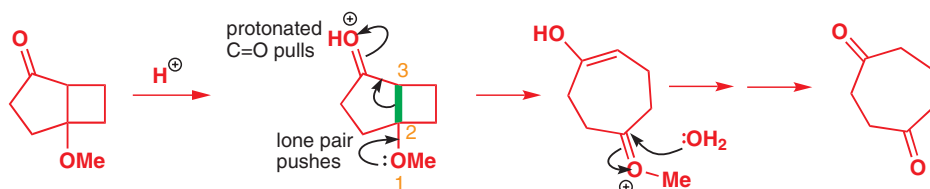
Protonated carbonyl compounds can be electron sinks too (remember the dienone-phenol rearrangement?) and this bicyclic methoxy ketone fragments to a seven-membered ring in acid. Note the same 1, 2, 3, 4 arrangement, with the bond between carbon atoms 2 and 3 fragmenting.

Bond	Typical bond energy, kJ mol ⁻¹
C–C	339
C–O	351
C–H	418
O–H	460

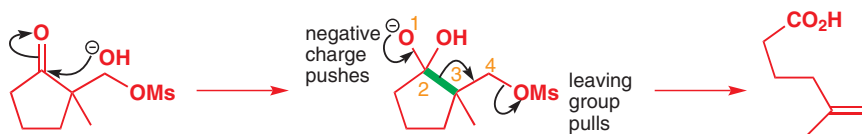
■ The bond energies listed in the table are the energies required to break the bonds **homolytically** to give two radicals, not **heterolytically** to give two ions. We will look at homolytic fragmentation in much more detail in the next chapter.

Interactive fragmentation mechanisms relying on bond polarization

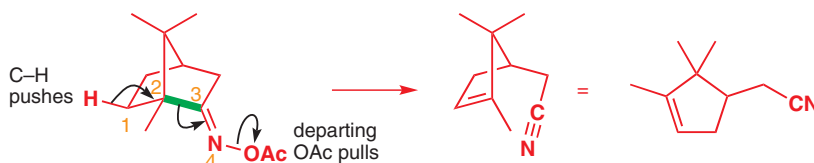
■ Note the numbering in these diagrams: 1, 2, 3, 4 from electron source to electron sink. We shall make use of it in many more fragmentation mechanisms.



A leaving group such as mesylate can exercise the ‘pull’ and in the next example a carbonyl group provides the ‘push’ after it has been attacked by a nucleophile. This five-membered cyclic ketone fragments on treatment with base—can you detect hints of the benzylic acid rearrangement?

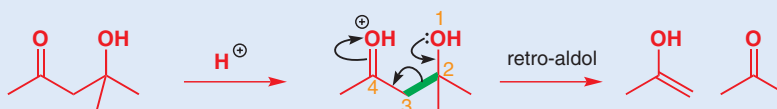


Analysing the Beckmann fragmentation on p. 960 in the same way, we can identify the electron sink (the departing acetate group), although the source in this case is a little more obscure. Saying that the tertiary cation is stable is really saying that the neighbouring C–C and C–H bonds provide electrons (through σ conjugation, see p. 334) to stabilize it, so these are the electron sources (the ‘push’). A good alternative is to write loss of a proton concerted with fragmentation, which gives one particular C–H bond as the source of electrons.



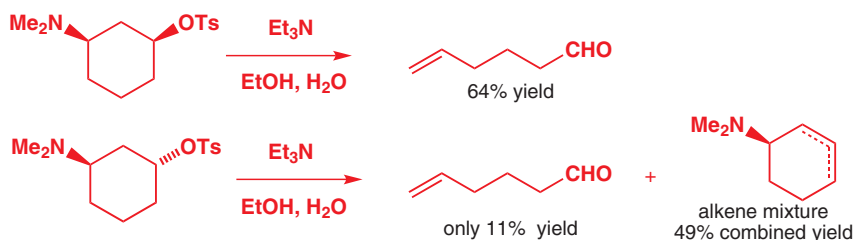
The retro-aldol is a fragmentation reaction

We should perhaps remind you here of the reversibility of the aldol reaction (Chapter 26): a retro-aldol is a fragmentation reaction with a carbonyl group as electron sink and OH as electron source. The aldol reaction usually goes in the other direction of course, but where steric or ring-strain factors are involved, this may not be the case.

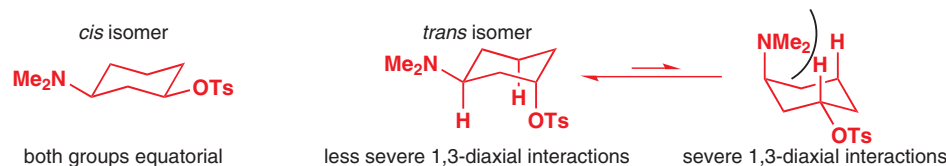


Fragmentations are controlled by stereochemistry

The control of rearrangements can be stereoelectronic in origin—if a molecule is to rearrange, orbitals have to be able to overlap. This means that, for a Beckmann *rearrangement*, the migrating group has to be *trans* to the leaving group. Not surprisingly, the same is true for Beckmann fragmentations like the one at the end of the last section, where the green fragmenting bond is *trans* to the leaving group. Before we extend these ideas any further, consider these two quite different reactions of quite similar compounds.



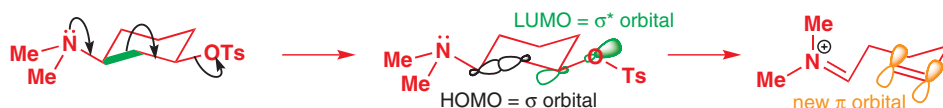
Just as with the rearrangements we looked at on p. 933, we need to draw these compounds in reasonable chair conformations in order to understand what is going on. In the *cis* isomer, both substituents can be equatorial; in the *trans* isomer one has to be axial, and this will be mainly the OTs group, since the two methyl groups of NMe₂ suffer greater 1,3-diaxial interactions.



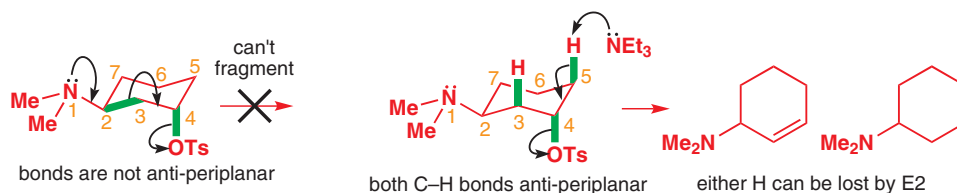
Now, the *cis* isomer has clearly undergone a fragmentation reaction and, as usual, numbering the atoms can help to identify the bond that breaks. The nitrogen lone pair pushes, the departing tosylate pulls, and the resulting iminium ion hydrolyses to the product aldehyde.



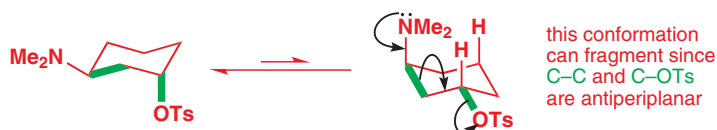
Yet the *trans* isomer does this only in very low yield. Mostly it eliminates TsOH to give a mixture of alkenes. Why? Well, notice that, in the *cis* isomer, the fragmenting bond is *trans* to the leaving group—indeed, it is both parallel and *trans* (in other words anti-periplanar) to the leaving group. Electrons can flow smoothly from the breaking σ bond into the σ^* of the C–OTs bond, forming as they do so a new π bond.



For the *trans* isomer, fragmentation of the most populated conformation is impossible because the leaving group is not anti-periplanar to any C–C bond. The only bonds anti-periplanar to OTs are C–H bonds, making this compound ideally set up for another reaction whose requirement for anti-periplanarity you have already met—E2 elimination.



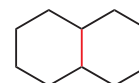
The other conformation can fragment because now the OTs is anti-periplanar to the right C–C bond, and this is probably where the 11% fragmentation product comes from.



Ring expansion by fragmentation

Ring sizes greater than eight are hard to make. Yet five- and six-membered rings are easy to make. Once you realize that a fused pair of six-membered rings is really a ten-membered ring with a bond across the middle, the potential for making medium rings by fragmentation becomes apparent.

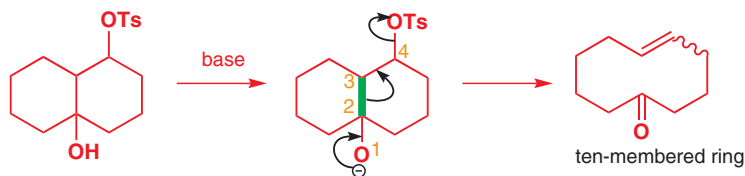
6,6-fused decalin



outer ten-membered ring

Interactive explanation of the stereochemistry of fragmentations

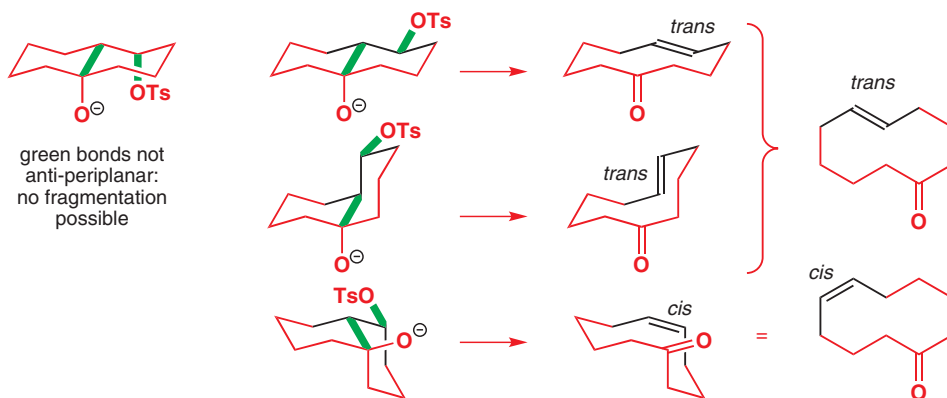
All you need to do is to make the bond to be broken the 2–3 bond in a 1, 2, 3, 4 electron source-sink arrangement and the ten-membered ring should appear out of the wreckage of the fragmentation. Here is an example:



Interactive mechanism for ring expansion by fragmentation

This is the simple overall result, but there is more to explore. The starting hydroxytosylate can exist as four diastereoisomers: two *trans*-decalins and two *cis*-decalins. What is more, the product has a double bond in a ten-membered ring: will it be *cis* or *trans*? (Both are possible in a ring with more than eight members: see Chapter 29.) One of the four diastereoisomers of the starting material cannot place the tosylate anti-periplanar to the ring-fusion bond, so it can't fragment. The other three diastereoisomers all can, but two of them give a *trans* double bond while the third gives *cis*.

➡ We discussed the conformations of decalins in Chapter 16.

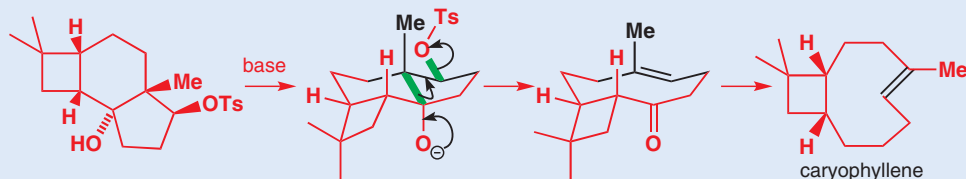
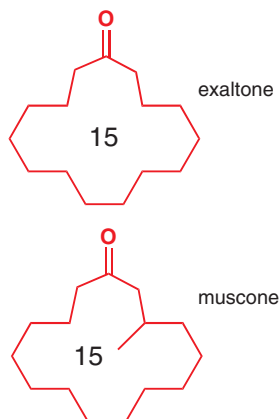


Interactive explanation of the importance of stereochemistry in ring expansions

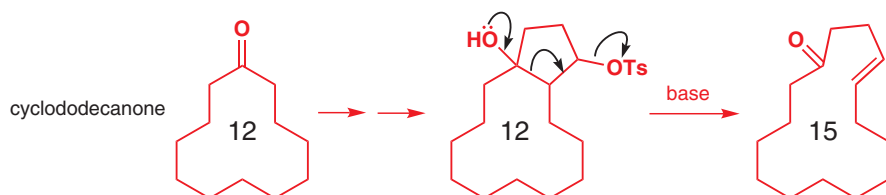
Looking at the alignment of the bonds that end up flanking the double bond in the product shows you where the geometrical isomers come from: these are the black bonds in the starting material, and are *trans* across the forming π system in the first two isomers and *cis* in the third. Fragmentations are stereospecific with regard to double-bond geometry, much as E2 elimination reactions are.

Caryophyllene by fragmentation

Corey applied this stereospecificity in conjunction with a ring expansion reaction to make the natural product caryophyllene. Caryophyllene is a bicyclic molecule with a nine-membered ring containing an *E* trisubstituted double bond. The right relative stereochemistry in the starting material leads both to fragmentation of the right bond and to formation of the alkene with the right stereochemistry.



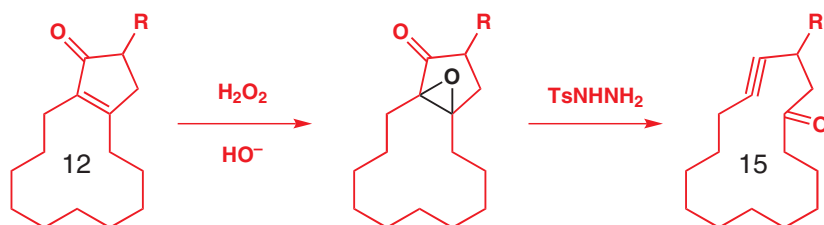
Muscone and exaltone are important perfumery compounds with even-harder-to-make 15-membered ring structures. Cyclododecanone is commercially available: addition of a fused five-membered ring and fragmentation of the 12,5-ring system is a useful route to these 15-membered ring compounds.



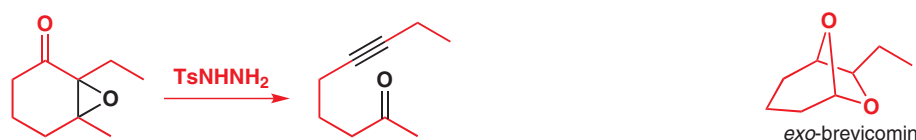
The Eschenmoser fragmentation

In the late 1960s, the Swiss chemist Albert Eschenmoser discovered an important reaction that can be used to achieve similar ring expansions and that now bears his name, the Eschenmoser fragmentation. The starting material for an Eschenmoser fragmentation is the epoxide of an α,β -unsaturated ketone. The fragmentation happens when this epoxyketone is treated with tosylhydrazine, and one of the remarkable things about the product is that it is an alkyne. The fragmentation happens across the epoxide (shown in black), and the product contains both a ketone (in a different place from the ketone in the starting material) and an alkyne. You can see how in this case hydrogenation of the triple bond can give muscone ($R=Me$) or exaltone ($R=H$).

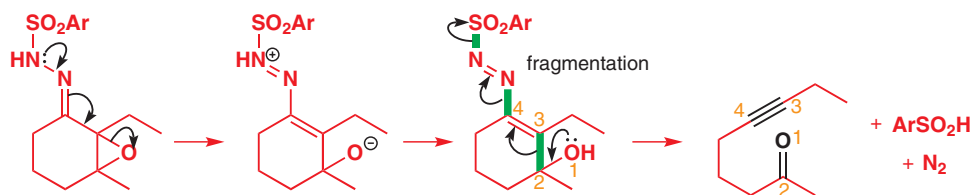
Albert Eschenmoser (1925–), working at the ETH in Zurich, synthesized vitamin B₁₂, at the time (1973) the most complicated molecule yet made, in what was for that era an unusual international collaboration with Woodward at Harvard.



The Eschenmoser fragmentation does not have to be a ring expansion, and it is a useful synthetic method for making keto-alkynes. The following reaction, which we will use to discuss the fragmentation's mechanism, was used to make an intermediate in the synthesis of an insect pheromone, *exo*-brevicomin.



The reaction starts with formation of the tosylhydrazone from the epoxyketone. The tosylhydrazone is unstable with respect to opening of the epoxide in an elimination reaction, and it is this elimination that sets up the familiar 1, 2, 3, 4 system ready for fragmentation. The 'push' comes from the newly created hydroxyl group, and the 'pull' from the irresistible concerted loss of a good leaving group (Ts^-) and an even better one (N_2). Notice how all the (green) bonds that break are parallel to one another, held anti-periplanar by two double bonds. Perfect!

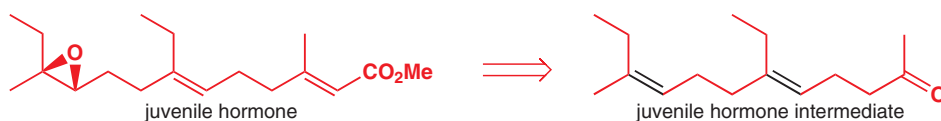


■ The epoxyketones are made by epoxidizing the electron-poor enones with basic hydrogen peroxide, see Chapter 22.

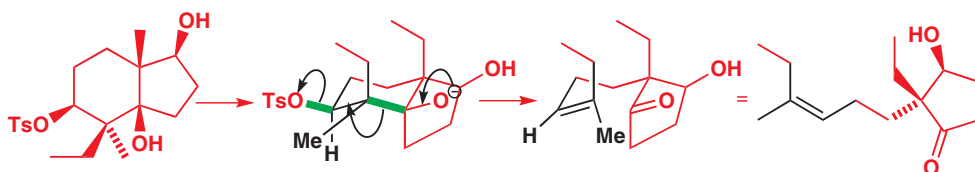
■ The sulfur-containing leaving group here is not toluenesulfonate (tosylate, or TsO^-) but toluenesulfinate ($ArSO_2^-$ or Ts^-), giving toluenesulfinic acid (TsH or $ArSO_2H$), not toluenesulfonic acid ($TsOH$ or $ArSO_3H$) as a by-product.

Controlling double bonds using fragmentation

Juvenile hormone (a compound you met in Chapter 27, p. 677) is a compound whose synthesis presents a major challenge: it requires the control of three trisubstituted double bonds (one of which ends up as an epoxide). The key intermediate shown contains two of them.

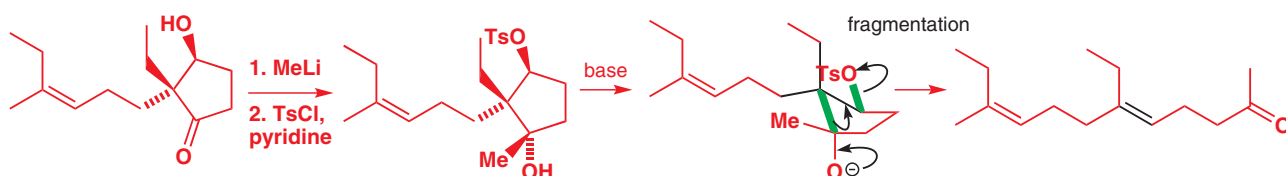


The chemists who succeeded in making this compound reasoned that, if this intermediate could be made stereospecifically by fragmenting a cyclic starting material, the (hard-to-control) double-bond stereochemistry would derive directly from the (easier-to-control) relative stereochemistry of the cyclic compound. The starting material they chose was a 5,6-fused system, which fragments to give one of the double bonds.



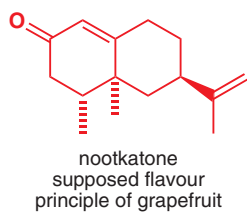
Interactive synthesis of juvenile hormone by fragmentations

The product of this reaction is prepared for another fragmentation by addition of methyl-lithium (you might like to consider why you get this diastereoisomer) and tosylation of the less hindered secondary alcohol. Base promotes the second fragmentation and gives the ketone with the two double bonds in place.



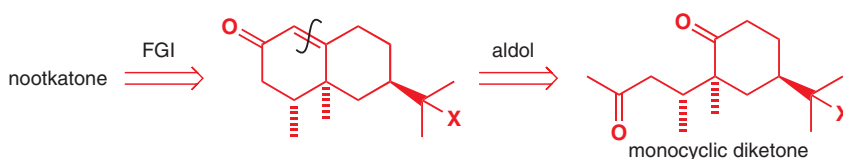
In the next chapter you will meet, among many other reactions, more fragmentations, but they will be radical fragmentations rather than ionic fragmentations, and involve homolytic cleavage of C–C bonds.

The synthesis of nootkatone: fragmentation showcase



To finish this chapter, we will present three different synthetic routes to the same compound, all of which illustrate the power of fragmentation in the synthesis of cyclic compounds. The story starts with grapefruit, which contains a simple bicyclic enone called nootkatone. It was assumed, wrongly as it happens, that the scent of grapefruit came from this compound, and in the 1970s there was quite a rush to synthesize this compound in various laboratories. A remarkable feature of many successful syntheses was the use of fragmentation reactions. We shall describe parts of three syntheses involving the fragmentation of a six-, a four-, and a three-membered ring.

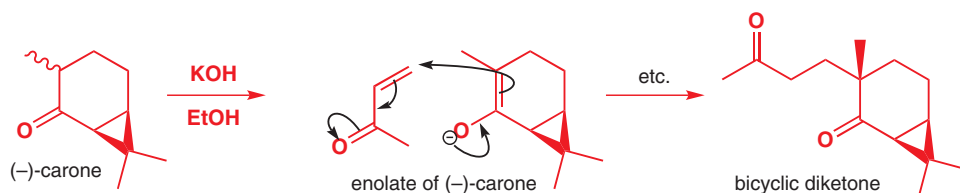
Most syntheses make the side-chain alkene by an elimination reaction so the first 'disconnection' is an FGI adding HX back into the alkene. The last C–C bond-forming operation in most syntheses is an intramolecular aldol reaction to make the enone so that can be disconnected next. It is the starting material for the aldol, a simple monocyclic diketone, which is usually made by a fragmentation reaction because this is a good way to set up the stereochemistry.



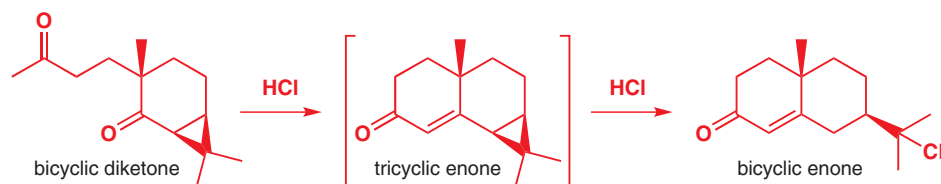
➡ The terminology ('disconnection', 'FGI') in this paragraph derives from Chapter 28.

Fragmentation of a three-membered ring

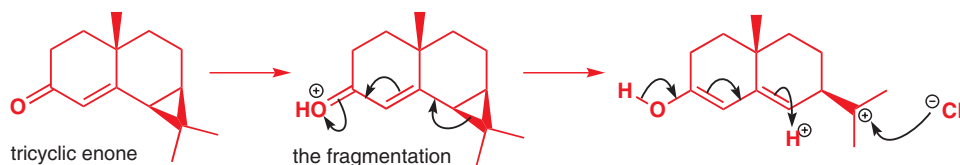
This synthesis does not look as though it will lead to nootkatone because the fragmentation product still requires a great deal of modification. It has the advantage that the stereochemistry is correct at one centre at least. The sequence starts from natural (–)-carone: conjugate addition of the enolate to butenone without control leads to a bicyclic diketone with one extra stereogenic centre. The enone adds to the bottom face of the enolate opposite the dimethylcyclopropane ring so the methyl group is forced upwards.



Now the diketone is cyclized by a Robinson-style aldol condensation in HCl to give a bicyclic enone. But during the reaction, a new six-membered ring has been formed while the old three-membered ring has disappeared, evidently by fragmentation.



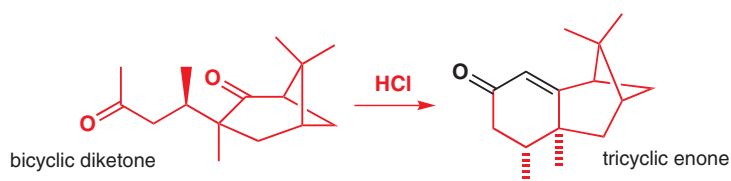
The fragmentation is pulled by the enone (with some help from the acid) and pushed by the stability of a tertiary carbocation as well as the release of strain as the single bond that is fragmented is in a three-membered ring.



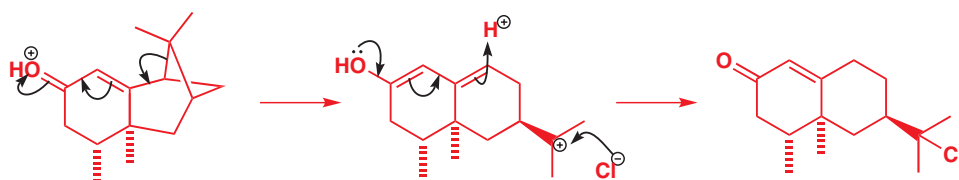
Addition of a proton to the end of the enol and a chloride ion to the cation gives the product. The further development of this compound into nootkatone is beyond the scope of this book.

Fragmentation of a four-membered ring

This approach leads directly to the enone needed for nootkatone. A diketone prepared from a natural terpene is also treated with HCl and much the same reactions ensue except that the fragmentation now breaks open a four-membered ring. First, the intramolecular aldol reaction to make the second six-membered ring.

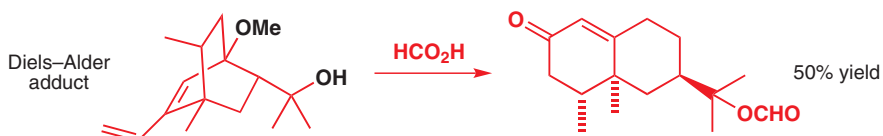


Now the fragmentation, which follows much the same course as the last one: the enone again provides the electron pull while the cleavage of a strained C–C single bond in a four-membered ring to give a tertiary carbocation provides the electron push. A simple elimination is all that is needed to make nootkatone from this bicyclic chloroenone.

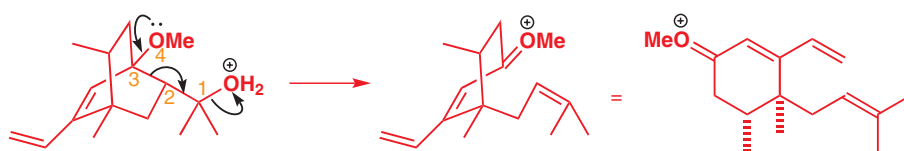


Fragmentation of a six-membered ring

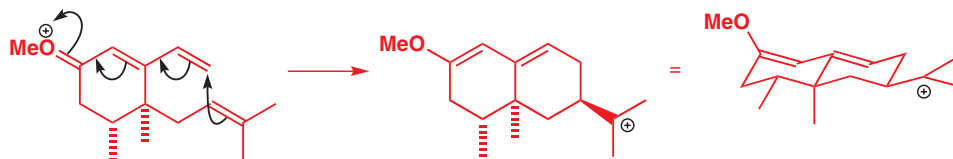
This chemistry is quite different from the examples we have just seen. The starting material has a bridged bicyclic structure and was made by a Diels–Alder reaction. Fragmentation is initiated by formic acid (HCO_2H), which protonates the tertiary alcohol and creates a tertiary carbocation. The ether provides the push. More serious electronic interactions are needed in this fragmentation as the C–C bond being broken is not in a strained ring.



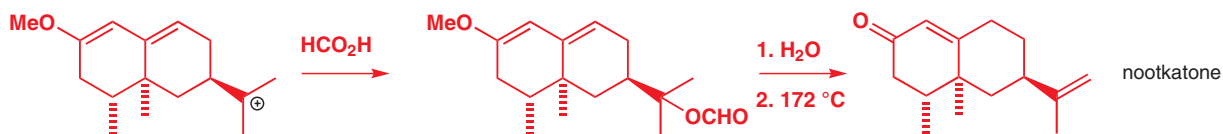
The yield of 50% may not seem wonderful, but there is obviously a lot of chemistry going on here so it is perfectly acceptable when so much is being achieved. The first stage is the fragmentation itself. Drawing the product first of all in the same shape as the starting material and then redrawing, to ensure that we don't make a mistake, we discover that we are well on the way to nootkatone. Note that the stereochemistry of the two methyl groups comes directly from the stereochemistry of the starting materials and no new stereogenic centres are created in the fragmentation. Although one six-membered ring is fragmented, another remains.



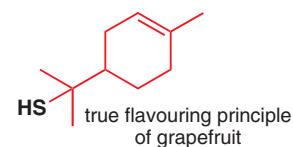
The first-formed product now cyclizes to form the second six-membered ring. This recreates a carbocation at the tertiary centre like the one that set off the fragmentation as the more nucleophilic end of the isolated alkene attacks the end of the conjugate electrophile. This is a thermodynamically controlled reaction with the new stereogenic centre choosing to have an equatorial substituent.



The cation picks up the only nucleophile available—formic acid. This gives the product of the fragmentation, which contains two unstable functional groups—a tertiary formate ester and an enol ether—and this product is not isolated from the reaction mixture. In water it hydrolyses to the enone, which undergoes elimination of formate to give nootkatone on heating.



Yet after all this effort, none of the synthetic samples of nootkatone delivered that intense grapefruit smell—for the simple reason that nootkatone is not the flavour principle of grapefruit! The samples of nootkatone that had been isolated from grapefruit contained minute traces of the true flavour principle—a simple thiol. Humans can detect 2×10^{-5} ppb (parts per billion) of this compound, so even the tiniest trace is very powerful. Nonetheless, the syntheses allowed chemists to correct a misconception.



Looking forward

Fragmentation reactions cleave C—C single bonds by a combination of electron push and electron pull so that both electrons in the bond move in the same direction as the bond breaks. In the next chapter we shall see reactions that break C—C bonds in a quite different way. No electron push or pull is required because one electron goes one way and one the other. These are radical reactions.

Further reading

F. A. Carey and R. J. Sundberg, *Advanced Organic Chemistry, Part A, Structure and Mechanisms*, Springer, 5th edn, 2007, part A. *Polar Rearrangements*, L. M. Harwood, Oxford Primer, OUP, 1992.

S. Warren and P. Wyatt, *Organic Synthesis: the Disconnection Approach*, 2nd edition, Wiley, Chichester, 2008, chapter 31. T.-L. Ho, *Heterolytic Fragmentation of Organic Molecules*, Wiley, 1993.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>